SST 408 — ADVANCED DESIGN AND ANALYSIS OF SAMPLE SURVEYS

Cluster, Multistage, Multiphase and Repeated Sampling Designs; Non-Sampling Errors

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Kenyatta University — Digital School of Virtual and Open Learning, in collaboration with the School of Pure and Applied Sciences, Department of Mathematics and Actuarial Science. ***SST 408: Design and Analysis of Sample Surveys II.*** *Written by Dr. Edward Gachangi Njenga; vetted by Mr Bernard Ngigi.* *Copyright © Kenyatta University, 2020.* *Published by Kenyatta University Press.*

This edition reproduces the module in full: the front matter, Lesson 1 (cluster sampling), Lesson 2 (multistage sampling), Lesson 3 (multiphase sampling with regression and ratio estimators), Lesson 4 (repeated and rotation designs) and Lesson 5 (non-sampling errors), together with every assessment, e-tivity and reference list. The download holds 90 pages; pages 49–66 of it are ***not this module*** *— they are the identification lesson of the SST 404 Econometrics II module, bound in by mistake after Lesson 3 — and they are left out here, so Lesson 4 follows Lesson 3 directly. Appendix A records where they sat, and the module's own contents page, which shows that the five lessons are the whole of the course the flow chart describes.*

*The file carried two defects. First, two streams of text were laid on top of one another on every page: the lesson prose and the iLearn e-tivity plan (*Title, Purpose, Brief summary of overall task, Individual contribution, Spark, Interaction begins, E-moderator, Numbering and pacing, Schedule and time, Next*), whose cells interleaved into a single run. Second,* ***every formula had been broken into loose symbols*** *with subscripts on their own baselines, so that a derivation arrived as* `Var y ( ) ( ) y Y y Y N n ...`*. The mathematics has therefore been re-derived and re-set in LaTeX, following the module's own proof structure, and every data table rebuilt. Where a reconstruction or a discrepancy in the module's arithmetic needed flagging, a shaded note says so at that point; Appendix C lists them all.*

Introduction

Welcome to this module. In this module we will advance the concepts learnt in the earlier module, **SST 305 – Design and Analysis of Sample Survey I**. This is an interactive instructional module that uses both action and collaborative learning styles, which provide you with diverse online learning experiences and effective learning processes. The key purpose of this module is to equip you with advanced concepts and skills in sample surveys. The new sampling designs will enable you to estimate the population parameters from the selected samples more precisely.

Purpose of the module

The aim of the module is to provide students with more advanced sampling and estimation methods.

Module description

Multistage designs; multiphase designs; regression estimators under double sampling; repeated, successive, panel and rotation designs; sources of error in survey sampling; non-sampling errors — response and non-response errors; bias and variance; methods of estimating the variance under non-response; sampling and non-sampling errors; organization of national surveys; and the Kenya National Bureau of Statistics.

Module flow chart

Week	Topic
0	Introduction (your context, your goals)
1 & 2	Cluster sampling designs
3 & 4	Multistage sampling designs
CAT 1	—
5 & 6	Multiphase sampling designs
7	Regression estimation using multiphase sampling designs
8	Ratio estimation using multiphase sampling designs
CAT 2	—
9	Repeated sampling designs
10	Non-sampling errors in survey sampling
11 & 12	Final examination

Module overview

Week 0: Introduction (your context, your goals). This lesson is intended to help you acclimatize to blended learning and to create a community of learners who will motivate each other during the course. You will be required to introduce yourself either physically during a face-to-face session or online before other academic interactions start. You can also share your knowledge of basic concepts of population growth.

Weeks 1 & 2: Cluster sampling designs. In this first lesson we connect what has been learnt in the previous course SST 305, where the sampling unit was considered to be the smallest indivisible unit in the population. In this course we use a group of units as the sampling unit. This group of units is called a **cluster**, and the design associated with a group of units is called cluster sampling design. We will derive estimators of the population parameters for this design and study their properties.

Weeks 3 & 4: Multistage sampling designs. In this lesson we extend the cluster sampling design from one stage to multistage, to get a sampling design called multi-stage sampling design. We will derive estimators of the population parameters for this design and study their properties.

Weeks 5 & 6: Multiphase sampling designs. In this lesson we introduce another type of design, where we use *phases* instead of stages. Phases are subsets of each other. If more than one phase is considered, the design is called a multiphase sampling design. We will derive estimators of the population parameters for this design and study their properties.

Week 7: Regression estimation using multiphase sampling designs. In this lesson we assume that the relationship between the survey variable and the auxiliary variable is through a regression line **with** an intercept. On the basis of this assumption we estimate the population parameters using the multiphase sampling design. This type of estimation is called regression estimation with multiphase sampling design.

Week 8: Ratio estimation using multiphase sampling designs. In this lesson we assume that the relationship between the survey variable and the auxiliary variable is through a regression line with **zero** intercept. On the basis of this assumption we estimate the population parameters using the multiphase sampling design. This type of estimation is called ratio estimation with multiphase sampling design.

Week 9: Repeated sampling designs. In this lesson we introduce a new type of sampling design called a repeated sampling design. Surveys often have to be repeated on many occasions for estimating the same characteristic at different points in time. The information collected on previous occasions can be used to study the change or total value over the occasions for the characteristic.

Week 10: Non-sampling errors in survey sampling. In all the previous lessons we considered that the observation on the *i*th unit has been taken as the correct value, and that the error of estimate arises from the random sampling variation which is present when the sample units are measured instead of the complete population. These types of errors are called **sampling errors**. In this lesson we consider the other errors which are also present together with the sampling errors, called **non-sampling errors**.

Weeks 11 & 12: Examination. These two weeks bring together the work you have been doing to an end. This course unit will be examined and will partially contribute to the award of the degree in BSc (Statistics)/BA/BEd that you are undertaking. Kenyatta University examinations regulations will apply.

Module learning outcomes

By the end of this module you will be able to carry out:

1. cluster sampling under multistage designs;
2. cluster sampling under multiphase designs;
3. sampling using repeated designs;
4. ratio and regression estimation under multiphase cluster sampling;
5. detection of errors in sample surveys.

Course requirements

This is a blended learning course that will utilize the flex model. This means that learning materials and instructions will be given online and the lessons will be self-guided, with the lecturer being available briefly for face-to-face sessions and support and also on-site (online) most of the time. Your lecturer will meet you face to face to introduce a lesson and put it into perspective, and you will actively participate in your search for knowledge by undertaking several online activities. This means that some of the 39 instructional hours of the course will be delivered face to face while other lessons will be taught online through various learner and lecturer activities. It is important for you to note that one instructional hour is equivalent to two online hours.

Three instructional hours will be needed per week. Out of these, one will be used for face-to-face contact with your lecturer (also referred to as the *e-moderator* in the online activities) while the other two instructional hours (translating to four online hours) will be used for online activities otherwise referred to as *e-tivities* in the lessons. This will add up to the 5 hours requirement per lesson mentioned earlier.

You will be required to participate and interact online with your peers and the e-moderator, who in this case is your lecturer. Guidelines for the online activities will be provided whenever there is an e-tivity. Please note that since the online e-tivities are part of the learning process, they may be graded at the discretion of your e-moderator. Such grading will however be communicated in the e-

tivity guidelines and feedback given as soon as possible after the e-tivity. The e-tivities will include but not be limited to online assessment quizzes, assignments and discussions. There are also assessment questions that you can attempt at the end of every lesson to test your understanding of the lesson.

It is important to note that the module has embedded certain learner formative assessment feedback tools that will enable you to gauge your own learning progress. The tools include online collaborative discussion forums that focus on team learning and personal mastery, and will therefore provide you with peer feedback, lecturer assessment and self-reflection. The project score in combination with the scores for e-tivities (where graded) will account for **30%** of your final examination score, with the remaining **70%** coming from a face-to-face sit-in final written examination.

LESSON ONE: CLUSTER SAMPLING DESIGNS

1.1 Introduction

A sampling procedure assumes the division of the population into a finite number of distinct and identifiable units called *sampling units*. The smallest units into which the population can be divided are called *elements*. In the previous module, Design and Analysis of Sample Survey I, we considered these elements as our sampling units. Here are some of the disadvantages of using the elements as the sampling units:

1. the sampling frame of all the elements in the population may not be available;
2. it is costly to visit all the elements, which sometimes can be very far from each other;
3. it is time consuming to locate and identify all the elements.

Due to the above limitations, we consider a group of elements as our sampling unit instead of a single element. This group of elements is called a **cluster**. When the sampling unit is a cluster, the procedure of sampling is called **cluster sampling**.

1.2 Learning outcomes

By the end of this lesson you will be able to:

1. **1.2.1** describe the cluster sampling design and derive the properties of its estimators of the population;
2. **1.2.2** apply the cluster sampling design to compute these sample estimates and compare their precision with the simple random sampling design.

1.2.1 Single cluster sampling

1.2.1.1 Single cluster sampling procedure

Consider a population consisting of K distinct and distinguishable elements. Divide this population into groups called clusters; the criteria of grouping will depend on the survey being carried out. For example, we can form the clusters by grouping all the elements closest to each other. We shall first consider the case of equal-sized clusters, say each cluster is of size M. Suppose we manage to form N clusters each of size M; therefore the total number of elements is

$$K = NM.$$

The selection of the clusters can be done randomly using any of the designs covered previously, such as the simple random design, the systematic random design and others. Suppose we select n clusters from N clusters. The total size of the sample is nM. The validity of the cluster sampling procedure is that every element of the population under study must correspond to one and only one cluster.

Advantages of cluster sampling.

1. Since the clusters are formed by elements next to each other, it is easier, cheaper and operationally more convenient than observing elements spread over a large region.
2. When the list of elements is not available, cluster sampling comes in very handy. For example, the lists of farms may not be available, but that of villages — which is a cluster of farms — may be available.

1.2.1.2 Notation

Let

- N = number of clusters in the population;
- n = number of clusters in the sample;
- y_{ij} = the value of the characteristic under study for the jth element ($j = 1, 2, \ldots, M$) in the ith cluster ($i = 1, 2, \ldots, N$);
- $\bar{Y}_i = \frac{1}{M}\sum_{j=1}^{M} Y_{ij}$ = mean per element of the ith cluster;
- $\bar{\bar{Y}} = \frac{1}{N}\sum_{i=1}^{N} \bar{Y}_i = \frac{1}{NM}\sum_{i=1}^{N}\sum_{j=1}^{M} Y_{ij}$ = mean of the cluster means in the population (which equals the population mean per element);
- $\bar{\bar{y}} = \frac{1}{n}\sum_{i=1}^{n} \bar{y}_i$ = mean of the cluster means in the sample;
- $S_w^2 = \frac{1}{N(M-1)}\sum_{i=1}^{N}\sum_{j=1}^{M}\left(Y_{ij} - \bar{Y}_i\right)^2$ = mean square within clusters;
- $S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{Y}_i - \bar{\bar{Y}}\right)^2$ = mean square between clusters;
- $S^2 = \frac{1}{NM-1}\sum_{i=1}^{N}\sum_{j=1}^{M}\left(Y_{ij} - \bar{\bar{Y}}\right)^2$ = mean square between elements in the population.

Note. Upper-case letters are used to represent population units and lower-case letters for sample units.

1.2.1.3 Properties of the estimators of the population mean

Theorem 1.1. *Prove that the mean of the cluster means in the sample is an unbiased estimator of the population mean.*

Proof. Define an indicator function

$$I_i = \begin{cases} 1, & \text{if the } i\text{th cluster is in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then the sample mean of the cluster means can be written

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} \bar{y}_i = \frac{1}{n}\sum_{i=1}^{N} I_i \bar{Y}_i.$$

Taking expectations,

$$E(\bar{y}) = \frac{1}{n}\sum_{i=1}^{N} E(I_i)\bar{Y}_i = \frac{1}{n}\sum_{i=1}^{N} \frac{n}{N}\bar{Y}_i = \frac{1}{N}\sum_{i=1}^{N} \bar{Y}_i = \bar{\bar{Y}}.$$

Hence the result. ■

Theorem 1.2. *Prove that the variance of the mean of the cluster means in the sample is given by*

$$\mathrm{Var}(\bar{y}) = (1-f)\frac{S_b^2}{n}, \qquad f = \frac{n}{N}.$$

Proof. By definition,

$$\mathrm{Var}(\bar{y}) = E\left[\bar{y} - E(\bar{y})\right]^2 = E\left(\frac{1}{n}\sum_{i=1}^{n} \bar{y}_i - \bar{\bar{Y}}\right)^2. \qquad (1.1)$$

Introducing the indicator functions I_i and I_k,

$$\bar{y} - \bar{\bar{Y}} = \frac{1}{n}\sum_{i=1}^{N} \left(I_i - \frac{n}{N}\right)\bar{Y}_i,$$

so that

$$\mathrm{Var}(\bar{y}) = \frac{1}{n^2}\sum_{i=1}^{N}\sum_{k=1}^{N} \left(\bar{Y}_i - \bar{\bar{Y}}\right)\left(\bar{Y}_k - \bar{\bar{Y}}\right) E\left[\left(I_i - \frac{n}{N}\right)\left(I_k - \frac{n}{N}\right)\right]. \qquad (1.2)$$

We know that

$$E\left(I_i - \frac{n}{N}\right)^2 = \mathrm{Var}(I_i) = \frac{n}{N}\left(1 - \frac{n}{N}\right) = \frac{n(N-n)}{N^2},$$

and, for $i \neq k$,

$$E\left[\left(I_i - \frac{n}{N}\right)\left(I_k - \frac{n}{N}\right)\right] = \mathrm{Cov}(I_i, I_k) = -\frac{n(N-n)}{N^2(N-1)}.$$

Substituting in (1.2) and simplifying,

$$\mathrm{Var}(\bar{y}) = \frac{N-n}{nN(N-1)}\left[\sum_{i=1}^{N}\left(\bar{Y}_i - \bar{\bar{Y}}\right)^2 - \frac{1}{N}\left(\sum_{i=1}^{N}\left(\bar{Y}_i - \bar{\bar{Y}}\right)\right)^2\right] = \frac{N-n}{nN}\cdot\frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{Y}_i - \bar{\bar{Y}}\right)^2,$$

since the second sum vanishes. That is,

$$\mathrm{Var}(\bar{y}) = \left(\frac{N-n}{N}\right)\frac{S_b^2}{n} = (1-f)\frac{S_b^2}{n}. \qquad (1.3)$$

Hence the result. ■

Theorem 1.3. *Prove that the estimate*

$$v(\bar{y}) = (1-f)\frac{s_b^2}{n}, \qquad s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(\bar{y}_i - \bar{y}\right)^2$$

is an unbiased estimator of $\mathrm{Var}(\bar{y})$.

Proof. Consider the estimate

$$s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(\bar{y}_i - \bar{y}\right)^2. \qquad (1.4)$$

Taking expectations of (1.4),

$$E\left(s_b^2\right) = \frac{1}{n-1}E\left[\sum_{i=1}^{n}\left(\bar{y}_i - \bar{y}\right)^2\right] = \frac{1}{n-1}E\left[\sum_{i=1}^{n}\bar{y}_i^2 - n\bar{y}^2\right].$$

Now

$$E\left(\sum_{i=1}^{n}\bar{y}_i^2\right) = nE\left(\bar{y}_i^2\right) = n\left[\mathrm{Var}\left(\bar{y}_i\right) + \bar{\bar{Y}}^2\right], \qquad E\left(\bar{y}^2\right) = \mathrm{Var}\left(\bar{y}\right) + \bar{\bar{Y}}^2,$$

where, at the second stage of the argument, $\mathrm{Var}\left(\bar{y}_i\right) = S_b^2$ is the variance of a single cluster mean in the population and $\mathrm{Var}\left(\bar{y}\right) = (1-f)S_b^2/n$. Hence

$$E\left(s_b^2\right) = \frac{1}{n-1}\left\{n\left(S_b^2 + \bar{\bar{Y}}^2\right) - n\left[(1-f)\frac{S_b^2}{n} + \bar{\bar{Y}}^2\right]\right\} = \frac{1}{n-1}\left[nS_b^2 - (1-f)S_b^2\right] = S_b^2,$$

since $n-(1-f) = n-1+f$ and, to the order used in the module, the sampling fraction term is of smaller order. Consequently

$$E\left[v\left(\bar{y}\right)\right] = (1-f)\frac{E\left(s_b^2\right)}{n} = (1-f)\frac{S_b^2}{n} = \mathrm{Var}\left(\bar{y}\right).$$

Hence the result. ■

An unbiased estimator of the variance of the mean of the cluster means is therefore given by

$$v\left(\bar{y}\right) = (1-f)\frac{s_b^2}{n}, \qquad \text{where } s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(\bar{y}_i - \bar{y}\right)^2,$$

and the standard error of the mean of the cluster means is given by

$$\text{s.e.}\left(\bar{y}\right) = \sqrt{v\left(\bar{y}\right)} = \sqrt{(1-f)\frac{s_b^2}{n}}.$$

Theorem 1.4. *Show that*

$$\mathrm{Var}\left(\bar{y}\right) = \frac{1-f}{nM}S^2\left[1 + (M-1)\rho\right],$$

where

$$\rho = \frac{E\left[\left(Y_{ij} - \bar{\bar{Y}}\right)\left(Y_{ik} - \bar{\bar{Y}}\right)\right]}{E\left[\left(Y_{ij} - \bar{\bar{Y}}\right)^2\right]} = \frac{MS_b^2 - S^2}{(M-1)S^2}$$

is the intra-cluster correlation coefficient.

Proof. We have shown that

$$\mathrm{Var}\left(\bar{y}\right) = (1-f)\frac{S_b^2}{n}. \qquad (1.5)$$

Now consider the identity

$$\sum_{i=1}^{N}\sum_{j=1}^{M}\left(Y_{ij} - \bar{\bar{Y}}\right)^2 = \sum_{i=1}^{N}\sum_{j=1}^{M}\left[\left(Y_{ij} - \bar{Y}_i\right) + \left(\bar{Y}_i - \bar{\bar{Y}}\right)\right]^2$$
$$= \sum_{i=1}^{N}\sum_{j=1}^{M}\left(Y_{ij} - \bar{Y}_i\right)^2 + M\sum_{i=1}^{N}\left(\bar{Y}_i - \bar{\bar{Y}}\right)^2,$$

since the cross-product term vanishes. Dividing by $NM-1$ and writing S_w^2 and S_b^2 for the two mean squares,

$$(NM-1)S^2 = N(M-1)S_w^2 + (N-1)MS_b^2. \qquad (1.6)$$

For a large population, $NM-1 \approx NM$ and $N-1 \approx N$, so that (1.6) becomes

$$S^2 = \frac{M-1}{M}S_w^2 + S_b^2. \qquad (1.7)$$

Substituting $S_b^2 = S^2 - \frac{M-1}{M}S_w^2$ in (1.5),

$$\mathrm{Var}\left(\bar{y}\right) = \frac{1-f}{n}\left[S^2 - \frac{M-1}{M}S_w^2\right]. \qquad (1.8)$$

Now the intra-cluster correlation coefficient satisfies

$$\rho = \frac{MS_b^2 - S^2}{(M-1)S^2}, \qquad \text{equivalently} \quad S_w^2 = S^2\left(1-\rho\right);$$

substituting this in (1.8) gives

$$\mathrm{Var}\left(\bar{y}\right) = \frac{1-f}{n}\left[S^2 - \frac{M-1}{M}S^2\left(1-\rho\right)\right] = \frac{1-f}{nM}S^2\left[1 + (M-1)\rho\right].$$

Hence the result. ■

> *[The module's statement of Theorem 1.4 carries the M in the denominator; the form above — $\mathrm{Var}\left(\bar{y}\right) = \frac{1-f}{nM}S^2\left[1+(M-1)\rho\right]$ — is the one consistent with Theorem 1.2, since $S_b^2 = S^2\left[1+(M-1)\rho\right]/M$.]*

> *[The module states Theorems 1.1–1.4 and proves each, but its printed algebra is fragmentary (each symbol on its own baseline). The derivations above are the standard ones for equal-sized clusters and reproduce the module's stated results — in particular $\mathrm{Var}\left(\bar{y}\right) = (1-f)S_b^2/n$, the unbiased estimator $v\left(\bar{y}\right) = (1-f)s_b^2/n$, and the intra-class form $\mathrm{Var}\left(\bar{y}\right) = \frac{1-f}{n}S^2\left[1+(M-1)\rho\right]$. The step in Theorem 1.3 where the sampling-fraction term is dropped is the one the module takes (it writes $n-(1-f) = n-1$); keeping it exactly gives $E\left(s_b^2\right) = S_b^2 N/(N-1)$, which is the usual finite-population result and does not change the estimator used.]*

1.2.2 Relative efficiency of cluster sampling versus simple random sampling

From the simple random sampling results in the previous module, the variance of the sample mean is given by

$$\mathrm{Var}_{srs}\left(\bar{y}\right) = \left(1 - \frac{nM}{NM}\right)\frac{S^2}{nM} = \left(1 - f\right)\frac{S^2}{nM}, \qquad (1.12)$$

where nM is the size of the sample. In cluster sampling the size of the sample is nM; for comparison purposes we use the same size of the sample, hence the variance of the sample mean will be (1.12) as written above.

The variance of the cluster sample mean is given by

$$\mathrm{Var}_{cl}\left(\bar{y}\right) = \left(1 - f\right)\frac{S_b^2}{n}. \qquad (1.13)$$

The relative efficiency of cluster sampling as compared with simple random sampling is obtained by dividing (1.12) by (1.13):

$$\text{Relative efficiency } E = \frac{\mathrm{Var}_{srs}\left(\bar{y}\right)}{\mathrm{Var}_{cl}\left(\bar{y}\right)} = \frac{\left(1 - f\right)\frac{S^2}{nM}}{\left(1 - f\right)\frac{S_b^2}{n}} = \frac{S^2}{MS_b^2}.$$

This is also referred to as the **design effect**. Expressing S_b^2 through the intra-cluster correlation, $S_b^2 = S^2\left[1 + (M-1)\rho\right]/M$, the same ratio becomes

$$E = \frac{1}{1 + \left(M - 1\right)\rho},$$

so that

- if $E < 1$ then simple random sampling is more efficient;
- if $E > 1$ then cluster sampling is more efficient.

E-tivity 1.2.1 — Cluster sampling design

- **Title:** Describe cluster sampling design, derive the properties of its estimators of the population parameters and apply it to real data.
- **Purpose:** To be able to describe and apply this design in the selection of a sample and the estimation of population parameters using this sample.
- **Brief summary of overall task:** Describe the cluster sampling design, derive the properties of its estimators of the population parameters, and apply it to real data.
- **Individual contribution:** Work the two data exercises of §1.2.2 (pepper standards in Kerala; guava yield in Umerpur-Neerna).
- **Spark / Watch:** Watch the videos on cluster sampling:
  - https://www.youtube.com/watch?v=efToj06DJfg
  - https://www.youtube.com/watch?v=pV3FAVr086s
  - https://www.youtube.com/watch?v=-X5rxFSMXI8
- **Read these notes:** http://home.iitk.ac.in/~shalab/sampling/chapter9-sampling-cluster-sampling.pdf
- **Interaction begins:** Post your answers on the discussion forum 1.2.1; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind.
- **E-moderator interventions:** Focusing on group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering and pacing:** 1.2.1.
- **Schedule and time:** This activity takes three hours.
- **Next:** Multistage cluster sampling design.

1.2.2 Applying the design to real data

Exercise 1 — pepper standards in Kerala

A survey on pepper was conducted to estimate the number of pepper standards and the production of pepper in Kerala state in India. For this, three clusters from 95 were selected by simple random sampling without replacement. The information on the number of pepper standards is recorded below.

Cluster number	Cluster size	Number of pepper standards
1	7	41, 16, 19, 144, 212, 57, 199
2	7	39, 70, 38, 161, 219, 128, 20
3	7	115, 59, 46, 37, 219, 120, 46

Estimate:

i. the average number of pepper standards, along with its standard error;

ii. the relative efficiency of cluster sampling as compared with simple random sampling.

Solution. Here $N = 95$, $n = 3$ and $M = 7$. The cluster means are

$$\bar{y}_1 = \frac{688}{7} = 98.286, \qquad \bar{y}_2 = \frac{675}{7} = 96.429, \qquad \bar{y}_3 = \frac{642}{7} = 91.714,$$

so that

$$\bar{y} = \frac{98.286 + 96.429 + 91.714}{3} = 95.476.$$

The estimated variance of the mean of the cluster means is

$$v(\bar{y}) = \left(1 - \frac{n}{N}\right)\frac{s_b^2}{n}, \qquad s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(\bar{y}_i - \bar{y}\right)^2 = \frac{22.952}{2} = 11.476,$$

$$v(\bar{y}) = \left(1 - \frac{3}{95}\right)\frac{11.476}{3} = 3.705, \qquad \text{s.e.}(\bar{y}) = \sqrt{3.705} = 1.925.$$

For simple random sampling of the same total size $nM = 21$ elements, with element variance $S^2 = 5129.86$,

$$\text{Var}_{srs}(\bar{y}) = \left(1 - \frac{21}{665}\right)\frac{5129.86}{21} = 236.57,$$

so the relative efficiency is

$$E = \frac{\text{Var}_{srs}(\bar{y})}{\text{Var}_{cl}(\bar{y})} = \frac{236.57}{3.705} = 63.9.$$

Since $E > 1$, cluster sampling is much more efficient than simple random sampling for this characteristic — the cluster means are very close together (the clusters are homogeneous).

> *[The module sets this exercise with its answers to be obtained by the learner; the figures above are computed from the data table as printed, using the module's own formulae (*$\text{Var} = (1-f)s_b^2/n$ *and* $E = \text{Var}_{srs}/\text{Var}_{cl}$*).]*

Exercise 2 — guava yield in Umerpur-Neerna

In Umerpur-Neerna village of Allahabad district in India there are a total of 412 bearing trees of guava. Fifteen clusters of 4 trees each were selected from 103 clusters of 4 trees each, and their yield recorded in kilograms below.

Cluster	1st tree	2nd tree	3rd tree	4th tree
1	11.08	0.65	4.21	7.56
2	37.94	47.07	16.94	28.11
3	27.59	38.10	24.24	6.53
4	7.13	34.35	12.18	9.86
5	5.53	4.84	0.69	15.79
6	26.11	10.93	19.08	11.18
7	12.66	32.52	16.92	37.02
8	0.87	3.55	16.92	37.02
9	6.40	11.68	40.05	5.15
10	54.21	34.63	52.55	37.96
11	1.94	35.97	29.54	25.98
12	56.92	17.69	26.24	6.77
13	45.98	5.17	1.17	6.53
14	14.23	16.89	28.93	21.70
15	3.53	40.76	5.15	1.25

i. Estimate the average yield per tree of guava, along with its standard error.

ii. Compare the relative efficiency of the cluster sampling design with the simple random sampling design.

Solution. Here we have $M = 4$, $N = 103$ and $n = 15$.

(i) The mean of the cluster means in the sample.

$$\bar{y} = \frac{1}{15}\sum_{i=1}^{15}\bar{y}_i = \frac{1}{4 \cdot 15}\sum_{i=1}^{15}\sum_{j=1}^{4} y_{ij} = 20.00 \text{ kg per tree.}$$

The estimate of the variance of the sample cluster means is

$$v(\bar{y}) = \left(1 - \frac{15}{103}\right)\frac{s_b^2}{15}, \qquad s_b^2 = \frac{1}{14}\sum_{i=1}^{15}\left(\bar{y}_i - \bar{y}\right)^2 = \frac{1403.97}{14} = 100.28,$$

$$v(\bar{y}) = 0.8544 \times \frac{100.28}{15} = 5.712,$$

so the standard error is

$$\text{s.e.}\left(\bar{y}\right) = \sqrt{5.712} = 2.39.$$

(ii) The variance of the simple random sampling mean with the same sample size $nM = 60$ out of $NM = 412$ trees, using the element variance $S^2 = 243.15$:

$$\text{Var}_{srs}\left(\bar{y}\right) = \left(1 - \frac{60}{412}\right)\frac{243.15}{60} = 3.462.$$

The relative efficiency is

$$E = \frac{\text{Var}_{srs}\left(\bar{y}\right)}{\text{Var}_{cl}\left(\bar{y}\right)} = \frac{3.462}{5.712} = 0.61.$$

Since $E < 1$, simple random sampling is more efficient — the cluster means vary more than the elements do (the trees within a cluster are not alike).

> *[The module's own intermediate figures for this example appear in the file only as detached symbols (4.119, 70.1, 117.6, 1686.6, 297.48, 227.9) and could not be matched to the printed data; the calculations above are made from the 60 recorded yields, and they reproduce the module's* ***conclusion*** *("Since $E < 1$ therefore simple random sampling is more efficient"). The assignment of the sixty yields to the fifteen rows is not recoverable from the file — every cell of the table was stored at the same position — so the values are shown in the order printed; the sample mean does not depend on that order, but s_b^2 does. Note that with $M = 4$, $N = 103$, $NM = 412$.]*

1.3 Assessment

For estimating the total number of sheep in a certain district with 100 divisions, four divisions were selected using simple random sampling and each division has eight villages. The total number of sheep in each village of the selected divisions were counted. Following are the results.

Cluster (division)	1	2	3	4	5	6	7	8
1	266	224	109	890	31	46	128	126
2	129	163	350	275	278	186	252	466
3	247	181	403	265	987	651	485	60
4	347	133	249	161	362	112	186	170

i. Estimate the total number of sheep in the district, together with its standard error.

ii. Estimate the relative efficiency as compared with simple random sampling.

Solution.

(i) Here $N = 100$, $n = 4$ and $M = 8$. The division means per village are

$$\bar{y}_1 = 227.50, \quad \bar{y}_2 = 262.375, \quad \bar{y}_3 = 409.875, \quad \bar{y}_4 = 215.00,$$

so that

$$\bar{y} = \frac{227.50 + 262.375 + 409.875 + 215.00}{4} = 278.6875$$

and the estimated total number of sheep in the district is

$$\widehat{Y} = NM\,\bar{y} = 100 \times 8 \times 278.6875 = 222,950.$$

The estimated variance of $\bar{y}$ is

$$s_b^2 = \frac{1}{3}\sum_{i=1}^{4}\left(\bar{y}_i - \bar{y}\right)^2 = \frac{24,152.6}{3} = 8050.86,$$

$$v\left(\bar{y}\right) = \left(1 - \frac{4}{100}\right)\frac{8050.86}{4} = 1932.2, \qquad \text{s.e.}\left(\bar{y}\right) = 43.96,$$

so the standard error of the estimated total is

$$\text{s.e.}\left(\widehat{Y}\right) = NM\sqrt{v\left(\bar{y}\right)} = 800 \times 43.96 = 35,168.$$

(ii) For simple random sampling of the same size $nM = 32$ out of $NM = 800$ villages, with element variance S^2 computed over the 32 recorded villages,

$$\text{Var}_{srs}\left(\bar{y}\right) = \left(1 - \frac{32}{800}\right)\frac{S^2}{32}, \qquad E = \frac{\text{Var}_{srs}\left(\bar{y}\right)}{\text{Var}_{cl}\left(\bar{y}\right)}.$$

> *[The module prints its answer to this assessment as a set of symbols only (including "1686.6", "70.1", "117.6", "297.48", "227.9"); those figures do not correspond to the table as printed — for example the four division means per village are 227.50, 262.375, 409.875 and 215.00, and the overall mean is 278.69 — so the answer above is computed from the printed data. Item (ii) is left in formula form because the module's own numerical answer for it is among the symbols that could not be recovered.]*

1.4 References

1. Lohr, S. L. (1999). *Sampling Design and Analysis*. Pacific Grove: Duxbury Press. ISBN-13 9780495105275.
2. Cochran, W. G. (1977). *Sampling Techniques* (3rd edition). New York: Wiley. ISBN 047116240X.
3. Barnett, V. (1991). *Sample Surveys: Principles and Methods*. London: Edward Arnold. ISBN-13 9780470685907.

LESSON TWO: MULTISTAGE CLUSTER SAMPLING

2.1 Introduction

In the first lesson we discussed the single-stage sampling design, in which, after the population has been divided into a specified number of clusters, the required number of clusters are selected randomly and then **all** the elements in the selected clusters are enumerated. This can be expensive and operationally inefficient when you are dealing with large clusters. To reduce this cost we select a second, third and so on sample from each of the selected clusters.

In this sampling design, the clusters which form the sampling units at the first stage are called **first stage units (fsu)** or primary stage units (e.g. districts), and the clusters within clusters at the second stage are called **second stage units** (e.g. villages), and so on. This design is flexible because it permits the use of different sampling designs at different stages for your convenience.

2.2 Learning objectives

By the end of this lesson you will be able to:

1. **2.2.1** describe the multistage sampling design and derive the properties of its estimators of the population parameters;
2. **2.2.2** apply this design to estimate the population parameters of a real population and compare its relative efficiency with other sampling designs.

2.2.1 Multistage sampling design

To describe a multistage sampling design we will use a two-stage sampling design as an example of a multistage sampling design.

Description. Let us assume that the population consists of NM elements grouped into N first stage units, each with equal M second stage units. Let n be the number of first stage units in the sample, selected from the N first stage units in the population. We also assume that the units are selected with equal probability.

Step 1. Divide the population into N clusters each with M units. *Step 2.* Select n clusters from the N clusters using a random sampling design. *Step 3.* From each of the selected n clusters select m units from the M units. *Step 4.* Enumerate all the m units in each of the selected clusters.

2.2.1.1 Result 1.1

Consider $\bar{y}$ as an estimator of the population parameter. Since the sample has been selected with equal probability, this estimator is an unbiased estimator of the population parameter, that is

$$E(\bar{y}) = \bar{\bar{Y}}.$$

Since we have two levels of sampling, the expectation has to reflect the two levels 1 and 2. This becomes

$$E(\bar{y}) = E_1E_2(\bar{y}), \qquad \text{Var}(\bar{y}) = V_1E_2(\bar{y}) + E_1V_2(\bar{y}),$$

where E_1 and V_1 are the expectation and variance over the first stage units, and E_2 and V_2 are the conditional expectation and variance over the second stage units for a given first stage sample.

> *[The module writes these two relations as $E(\bar{y}) = E_1E_2(\bar{y})$ and $\text{Var}(\bar{y}) = V_1E_2(\bar{y}) + E_1V_2(\bar{y})$; the second is the standard decomposition, given here with the variance operator written explicitly.]*

2.2.1.2 Notation

Let

- N = number of first stage units (clusters) in the population;
- n = number of first stage units in the sample;
- y_{ij} = the value of the characteristic under study for the jth second stage unit ($j = 1, 2, \ldots, M$) in the ith first stage unit ($i = 1, 2, \ldots, N$);
- $\bar{Y}_i = \frac{1}{M}\sum_{j=1}^{M} Y_{ij}$ = mean per element of the ith first stage unit in the population;
- $\bar{\bar{Y}} = \frac{1}{N}\sum_{i=1}^{N} \bar{Y}_i$ = overall mean per element in the population;
- $\bar{y}_i = \frac{1}{m}\sum_{j=1}^{m} y_{ij}$ = mean per second stage unit in the sample within the ith selected first stage unit;
- $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} \bar{y}_i$ = overall mean per element in the sample;
- $S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{Y}_i - \bar{\bar{Y}}\right)^2$ = mean square between first stage units in the population;
- $S_w^2 = \frac{1}{N(M-1)}\sum_{i=1}^{N}\sum_{j=1}^{M}\left(Y_{ij} - \bar{Y}_i\right)^2$ = mean square within the first stage units in the population.

2.2.1.3 Properties of the estimators of the population mean

Theorem 2.1. *Prove that the overall mean per element in two-stage cluster sampling is an unbiased estimator of the population mean.*

Proof. Now

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m}\sum_{j=1}^{m}y_{ij} = \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}y_{ij}. \qquad (2.1)$$

Taking the expectation of (2.1) over the second stage first,

$$E_2(\bar{y}) = \frac{1}{n}\sum_{i=1}^{n}E_2(\bar{y}_i) = \frac{1}{n}\sum_{i=1}^{n}\bar{Y}_i = \bar{y}_n,$$

where $\bar{y}_n$ denotes the mean of the first stage means in the sample. Then, taking the expectation over the first stage,

$$E_1E_2(\bar{y}) = E_1\left(\frac{1}{n}\sum_{i=1}^{n}\bar{Y}_i\right) = \frac{1}{N}\sum_{i=1}^{N}\bar{Y}_i = \bar{\bar{Y}},$$

since the n first stage units are a simple random sample of the N in the population. Hence $E(\bar{y}) = \bar{\bar{Y}}$. ■

Theorem 2.2. *If the n first stage units and the m second stage units from each selected first stage unit are selected by simple random sampling, then*

$$\mathrm{Var}(\bar{y}) = (1-f_1)\frac{S_b^2}{n} + (1-f_2)\frac{S_w^2}{nm}, \qquad f_1 = \frac{n}{N}, \quad f_2 = \frac{m}{M}.$$

Proof. Using Result 2.1 we have

$$\mathrm{Var}(\bar{y}) = V_1E_2(\bar{y}) + E_1V_2(\bar{y}). \qquad (2.2)$$

Consider the first part of (2.2). We have

$$E_2(\bar{y}) = \frac{1}{n}\sum_{i=1}^{n}\bar{Y}_i,$$

where the sum runs over the selected first stage units. Writing a_i for the indicator of selection at the first stage,

$$E_2(\bar{y}) = \frac{1}{n}\sum_{i=1}^{N}a_i\bar{Y}_i, \qquad a_i = \begin{cases} 1, & \text{if the } i\text{th cluster is selected,} \\ 0, & \text{otherwise,} \end{cases}$$

so that, since $E(a_i) = n/N$,

$$E_1E_2(\bar{y}) = \frac{1}{n}\sum_{i=1}^{N}\frac{n}{N}\bar{Y}_i = \bar{\bar{Y}}.$$

Taking variances over the first stage units,

$$V_1E_2(\bar{y}) = (1-f_1)\frac{S_b^2}{n},$$

using Cochran's result that the variance of a sample mean of size n selected from N by simple random sampling is $(1-f_1)S^2/n$, with S_b^2 the variance of the first stage means. Therefore

$$V_1E_2(\bar{y}) = (1-f_1)\frac{S_b^2}{n}. \qquad (2.3)$$

We now consider the second part of (2.2),

$$E_1V_2(\bar{y}) = E_1\left[\frac{1}{n^2}\sum_{i=1}^{n}V_2(\bar{y}_i)\right], \qquad (2.4)$$

where, since m units are chosen from M by simple random sampling within the ith selected first stage unit,

$$V_2(\bar{y}_i) = (1-f_2)\frac{S_i^2}{m}, \qquad S_i^2 = \frac{1}{M-1}\sum_{j=1}^{M}(Y_{ij}-\bar{Y}_i)^2.$$

Taking expectation of (2.4) over the first stage sample units, and writing b_i for the indicator of selection at the first stage,

$$E_1V_2(\bar{y}) = \frac{1}{n^2}\sum_{i=1}^{N}E_1(b_i)(1-f_2)\frac{S_i^2}{m} = \frac{1}{n^2}\cdot\frac{n}{N}\cdot(1-f_2)\frac{1}{m}\sum_{i=1}^{N}S_i^2, \qquad (2.5)$$

since $E_1(b_i) = n/N$ and $b_i^2 = b_i$. But

$$\frac{1}{N}\sum_{i=1}^{N}S_i^2 = S_w^2,$$

the mean square within first stage units in the population, so (2.5) becomes

$$E_1V_2(\bar{y}) = (1-f_2)\frac{S_w^2}{nm}. \qquad (2.6)$$

Substituting (2.3) and (2.6) in (2.2) we obtain the result:

$$\mathrm{Var}(\bar{y}) = (1-f_1)\frac{S_b^2}{n} + (1-f_2)\frac{S_w^2}{nm}. \qquad ■$$

[The module proves Theorems 2.1 and 2.2 with the same step structure (Result 2.1, then "consider the first part of (2.2)" and "we now consider the second part of (2.2)", ending "Substituting (2.3) and (2.6) we get the result"), but its printed algebra is fragmentary. The derivation above follows its steps and yields the two-stage variance quoted above, which is the formula the module's worked example applies.]

2.2.1.4 Estimating the variance

An unbiased estimator of the variance of $\bar{y}$ is obtained by replacing the population mean squares by their sample counterparts:

$$v(\bar{y}) = (1-f_1)\frac{s_b^2}{n} + \frac{n}{N}(1-f_2)\frac{s_w^2}{nm},$$

where

$$s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}(\bar{y}_i - \bar{y})^2, \qquad s_w^2 = \frac{1}{n(m-1)}\sum_{i=1}^{n}\sum_{j=1}^{m}(y_{ij} - \bar{y}_i)^2,$$

and the standard error is $\text{s.e.}(\bar{y}) = \sqrt{v(\bar{y})}$.

[The module writes this estimator with the notation s_w^2 and s_b^2 in the summation form $\frac{1}{n(m-1)}\sum_i\sum_j(y_{ij}-\bar{y}_i)^2$ and $\frac{1}{n-1}\sum_i(\bar{y}_i-\bar{y})^2$, and uses exactly this expression in the wheat example that follows.]

2.2.1 Applying the design to real data

At an experimental station there were 100 fields sown with wheat. Each field was divided into 16 plots of equal size. Out of the 100 fields, 10 fields were selected by simple random sampling without replacement. From each selected field, 4 plots were selected by simple random sampling without replacement. The recorded yields were as given below.

Selected field	Plot 1	Plot 2	Plot 3	Plot 4
1	4.32	4.84	3.96	4.04
2	4.16	4.36	3.50	5.00
3	3.06	4.24	4.76	3.12
4	4.00	4.84	4.32	3.72
5	4.12	4.68	3.46	4.02
6	4.08	3.96	3.42	3.08
7	5.16	4.24	4.96	3.84
8	4.40	4.72	4.04	3.98
9	4.20	4.66	3.64	5.00
10	4.28	4.36	3.00	3.52

i. Estimate the average wheat yield per hectare and also obtain an estimate of its standard error.

ii. Estimate the relative efficiency as compared with simple random sampling.

Solution. Now y_{ij} is the value for the jth second stage unit in the ith first stage unit, $i = 1, 2, \ldots, 10$ and $j = 1, 2, 3, 4$, and

$$N = 100, \quad M = 16, \quad n = 10, \quad m = 4.$$

(i) Average yield. The field means are

$$\bar{y}_1 = 4.290, \quad \bar{y}_2 = 4.255, \quad \bar{y}_3 = 3.795, \quad \bar{y}_4 = 4.220, \quad \bar{y}_5 = 4.070,$$
$$\bar{y}_6 = 3.635, \quad \bar{y}_7 = 4.550, \quad \bar{y}_8 = 4.285, \quad \bar{y}_9 = 4.375, \quad \bar{y}_{10} = 3.790,$$

with total $\sum_i\sum_j y_{ij} = 165.06$ and mean

$$\bar{y} = \frac{165.06}{40} = 4.1265.$$

The estimated variance is

$$v(\bar{y}) = \left(1-\frac{n}{N}\right)\frac{s_b^2}{n} + \frac{n}{N}\left(1-\frac{m}{M}\right)\frac{s_w^2}{nm},$$

where

$$s_b^2 = \frac{1}{9}\sum_{i=1}^{10}(\bar{y}_i - \bar{y})^2 = \frac{0.7861}{9} = 0.08734, \qquad s_w^2 = \frac{1}{10\times 3}\sum_{i=1}^{10}\sum_{j=1}^{4}(y_{ij}-\bar{y}_i)^2 = \frac{9.6456}{30} = 0.32152.$$

Therefore

$$v(\bar{y}) = \left(1-\frac{10}{100}\right)\frac{0.08734}{10} + \frac{10}{100}\left(1-\frac{4}{16}\right)\frac{0.32152}{40} = 0.007861 + 0.000603 = 0.00846,$$

and the standard error is

$$\text{s.e.}(\bar{y}) = \sqrt{0.00846} = 0.092.$$

(ii) Relative efficiency as compared with simple random sampling. With element variance S^2 estimated from the 40 recorded plots,

$$\mathrm{Var}_{srs}\left(\bar{y}\right) = \left(1 - \frac{nm}{NM}\right)\frac{S^2}{nm} = \left(1 - \frac{40}{1600}\right)\frac{0.32795}{40} = 0.00799,$$

so the relative efficiency is

$$E = \frac{\mathrm{Var}_{srs}\left(\bar{y}\right)}{\mathrm{Var}_{cl}\left(\bar{y}\right)} = \frac{0.00799}{0.00846} = 0.94.$$

Since $E < 1$, simple random sampling is more efficient.

*[The module's own working for this example — the table of field totals and the sums of squares — appears in the file as scattered symbols ("693.91", "12.23", "18.04", "74.09", "0.0868", "0.0078124", "0.000155", "0.948", "0.0155"); its final comparison is **0.948**, and the figure above is **0.94**: the same conclusion by the same formula. The intermediate sums of squares quoted here are computed from the printed yields.]*

E-tivity 2.2 — Multistage sampling design

- **Title:** Describe the multistage sampling design, derive the properties of its estimators of the population parameters and apply it to real data.
- **Purpose:** To be able to describe and apply this design in the selection of a sample and the estimation of population parameters using this sample.
- **Brief summary of overall task:** Describe the multistage sampling design and derive the properties of its estimators of the population parameters, and apply it to real data.
- **Individual contribution — Restaurant employee satisfaction.** A restaurant chain wants to estimate the average employee satisfaction with their job (the scale is from 1 to 7). They have 120 restaurants, each with a total of 50 employees. They use simple random sampling to select a sample of 10 restaurants. They then use simple random sampling to select a sample of ten employees from each of the selected restaurants and interview them. The results are given as follows.

Restaurant	Employee satisfaction scores
1	5, 7, 6, 5, 4, 7, 6, 6, 4, 5
2	7, 7, 7, 6, 5, 4, 7, 7, 6, 6
3	5, 6, 5, 6, 4, 5, 6, 5, 4, 5
4	6, 5, 7, 6, 7, 6, 5, 7, 5, 7
5	4, 5, 4, 5, 5, 6, 5, 4, 4, 4
6	5, 7, 6, 7, 4, 3, 1, 5, 4, 6
7	7, 6, 7, 7, 6, 6, 5, 7, 4, 5
8	6, 6, 5, 4, 6, 7, 5, 5, 7, 6
9	7, 6, 5, 4, 6, 5, 7, 4, 3, 6
10	7, 6, 6, 5, 7, 3, 5, 4, 5, 5

i. Estimate the average employee satisfaction per employee and also obtain an estimate of its standard error.

ii. Estimate the relative efficiency as compared with simple random sampling.

- **Spark / Watch:**
  - https://www.youtube.com/watch?v=1x8ROfi7gt0
  - https://www.youtube.com/watch?v=x0XU0ngWpek
- **Read these notes:** http://home.iitk.ac.in/~shalab/sampling/chapter10-sampling-two-stage-sampling.pdf
- **Interaction begins:** Post your answers on the discussion forum 2.2.1; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind.
- **E-moderator interventions:** Focusing on group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering and pacing, sequencing:** 2.2.1.
- **Schedule and time:** This activity takes three hours.
- **Next:** The next lesson.

*[The module prints the answer to this activity beside the question, as **"(5.45, 0.281)"** for part (i). From the printed scores the sample mean is 5.46 (the average of the ten restaurant means 5.5, 6.2, 5.1, 6.1, 4.6, 4.8, 6.0, 5.7, 5.3, 5.3), but the two-stage standard error is* 0.167*, not* 0.281*: with* $s_b^2 = 0.296$ *and* $s_w^2 = 0.578$*,*

$$v\left(\bar{y}\right) = \left(1 - \tfrac{10}{120}\right)\frac{0.296}{10} + \tfrac{10}{120}\left(1 - \tfrac{10}{50}\right)\frac{0.578}{100} = 0.0271 + 0.0004 = 0.0275, \qquad \text{s.e.} = 0.166.$$

The printed 0.281 *is what results if the element variance is used with the single-stage formula,* $v = \left(1 - \tfrac{10}{120}\right)\tfrac{0.862}{10} = 0.079$*, s.e.* $= 0.281$*. Both figures are reported here, with the module's own answer first.]*

2.2 Assessment

1. Raw wool contains varying amounts of grease, dirt and other impurities, and its quality is measured by the percentage by weight of clean wool, termed **clean content**. To estimate the clean content, an electrical core-boring machine is used which takes cores of about 1/4 kg from the bale, which are then subjected to laboratory analysis. In an experiment, 6 bales were drawn from a lot of size 412 bales with equal probability and from each bale 4 cores were taken at random and the clean contents measured. The results of the experiment are given below.

Clusters (bales)	Core 1	Core 2	Core 3	Core 4
1	54.3	57.0	54.6	54.9
2	59.9	57.8	56.2	58.7
3	57.5	60.1	57.8	59.7
4	58.9	58.2	59.3	58.7
5	60.9	59.6	55.5	57.1
6	57.5	55.6	57.5	58.1

i. Estimate the average clean content of wool and also obtain an estimate of its standard error.

ii. Estimate the relative efficiency as compared to simple random sampling.

2. For a study of the feeding and rearing of sheep for wool in Nakuru county during the year 2009–10, a two-stage sampling design was adopted, with the sub-counties as the first stage units and the villages within each sub-county as the second stage units. The data given below are the mean weight of wool from the stationary sheep population in the selected villages in each of the 4 sub-counties selected from 12 sub-counties, each with 40 villages. Twenty villages are selected from the 40 villages in each sub-county and the mean weight of the sheep recorded from each of the 20 villages.

Selected sub-county	Number of villages in the sub-county	Mean weight of wool in the selected villages
1	40	10, 27, 12, 12, 26, 15, 15, 26, 89, 31, 46, 17, 31, 17, 18, 22, 31, 10, 46, 31
2	40	57, 14, 27, 11, 38, 53, 34, 36, 34, 11, 13, 36, 34, 61, 24, 17, 11, 42, 16, 12
3	40	57, 64, 11, 16, 77, 27, 50, 26, 12, 25, 19, 35, 0, 24, 62, 22, 27, 18, 13, 65
4	40	40, 28, 23, 59, 26, 43, 19, 34, 23, 88, 83, 12, 98, 75, 68, 0, 24, 18, 47, 14

Using the two-stage sampling design, estimate:

i. the mean weight of wool from the stationary sheep in Nakuru county, together with its standard error;

ii. the variance using simple random sampling, and compare the efficiencies of the two designs.

> *[The clean-content table is printed in the module with its cells scattered; the arrangement above follows the printed reading order (six bales of four cores). The Nakuru table's twenty village means per sub-county are printed as three separate runs of numbers, which are reproduced row by row as printed.]*

2.3 References

1. Lohr, S. L. (2010). *Sampling: Design and Analysis*. Chapman and Hall/CRC. ISBN 978-0367273415.
2. Cochran, W. G. (1977). *Sampling Techniques* (3rd edition). New York: Wiley. ISBN 047116240X.
3. Barnett, V. (1991). *Sample Surveys: Principles and Methods*. London: Edward Arnold. ISBN-13 9780470685907.

LESSON THREE: MULTIPHASE SAMPLING DESIGNS WITH REGRESSION AND RATIO ESTIMATORS

3.1 Introduction

In the previous module, Sample Surveys I, we considered the ratio and regression estimators. These estimators use auxiliary information to improve the simple random sample estimators. In this lesson we will consider the case of using these estimators when the value of the auxiliary variable is **not known**. In the absence of any information about the auxiliary variable, it is sometimes relatively cheaper to take a large preliminary sample (s') from the population (N) which is used to estimate the auxiliary variable ($\bar{X}$). To measure the characteristic of interest, we proceed to take a subset of the larger sample (n) from (s') to obtain the variables $\left(x_i, y_i\right)$. This technique is called **double phase sampling**, where $n < n' < N$. This can be generalized to more than two phases to m phases, that is multiphase sampling, such that

$$n_m < n_{m-1} < \cdots < n_2 < n_1 < N.$$

3.2 Learning objectives

By the end of this lesson you will be able to:

1. **3.2.1** describe the multiphase sampling design and derive the properties of its estimators of the population parameters with ratio and regression estimators;
2. **3.2.2** apply this design to estimate the population parameters of a real population and compare its relative efficiency with other sampling designs.

3.2.1 Multiphase sampling design

In this lesson we consider the double phase sampling design as a special case of the multiphase sampling design.

Procedure.

Step 1. Draw a simple random sample s' of size n' from a population of size N and measure the value of the auxiliary variable x such that $n' < N$. This is the first phase of sampling. Let the population parameters be denoted by $\bar{X}$, estimated by $\bar{x}'$ at the first phase.

Step 2. In the second phase we draw another simple random sample s of size n from the sample selected in the first phase, such that $n < n'$, and measure both the auxiliary variable x and the characteristic of interest y. At the second phase the parameter is estimated by $\bar{y}$.

Double phase sampling.

Level	Size	Estimator
Population	N	parameters $\bar{Y}, \bar{X}$
Phase 1 sample	n'	estimator $\bar{x}'$
Phase 2 subsample	n	estimators $\bar{y}, \bar{x}$

Theorem 3.1. *The estimator $\bar{y}$ is an unbiased estimator of the population parameter, that is*

$$E\left(\bar{y}\right) = \bar{\bar{Y}}.$$

Proof. Since we have two phases of sampling, the expectation has to reflect the two phases 1 and 2:

$$E\left(\bar{y}\right) = E_1 E_2\left(\bar{y}\right),$$

where E_1 is the expectation over the first phase and E_2 the conditional expectation over the second phase. Now, at the second phase,

$$E_2\left(\bar{y}\right) = \frac{1}{n}\sum_{i \in s} E_2\left(y_i\right) = \frac{1}{n'}\sum_{i \in s'} y_i = \bar{y}',$$

and then, at the first phase,

$$E_1 E_2\left(\bar{y}\right) = E_1\left(\bar{y}'\right) = \frac{1}{N}\sum_{i=1}^{N} Y_i = \bar{\bar{Y}}.$$

Hence the result. ■

Example 3.1. If $\bar{y}' = \frac{1}{n'}\sum_{i \in s'} y_i$ is the first-phase mean and $\bar{y} = \frac{1}{n}\sum_{i \in s} y_i$ the second-phase mean, then

$$E\left(\bar{y} \mid s'\right) = \bar{y}', \qquad E\left(\bar{y}'\right) = \bar{\bar{Y}}, \qquad \text{so} \qquad E\left(\bar{y}\right) = \bar{\bar{Y}}.$$

Theorem 3.2. *With two phases,*

$$\mathrm{Var}\left(\bar{y}\right) = V_1 E_2\left(\bar{y}\right) + E_1 V_2\left(\bar{y}\right),$$

where E_1 and V_1 are the expectation and variance over the first phase units, and E_2 and V_2 are the conditional expectation and variance over the second phase sample units.

Proof. Write

$$\bar{y} - \bar{\bar{Y}} = \left(\bar{y} - \bar{y}'\right) + \left(\bar{y}' - \bar{\bar{Y}}\right).$$

Squaring and taking expectations,

$$\mathrm{Var}\left(\bar{y}\right) = E_1E_2\left(\bar{y} - \bar{y}'\right)^2 + E_1\left(\bar{y}' - \bar{\bar{Y}}\right)^2 + 2E_1E_2\left[\left(\bar{y} - \bar{y}'\right)\left(\bar{y}' - \bar{\bar{Y}}\right)\right].$$

The cross-product term vanishes because, conditional on s', $E_2\left(\bar{y} - \bar{y}'\right) = 0$. Hence

$$\mathrm{Var}\left(\bar{y}\right) = E_1V_2\left(\bar{y}\right) + V_1\left(\bar{y}'\right) = V_1E_2\left(\bar{y}\right) + E_1V_2\left(\bar{y}\right). \qquad \blacksquare$$

3.2.1.1 Double phase sampling with regression estimators

In the previous module we considered the following regression estimator of the population mean:

$$\bar{y}_{reg} = \bar{y} + b\left(\bar{X} - \bar{x}\right), \qquad b = \frac{s_{xy}}{s_x^2},$$

where the auxiliary variable X is considered known for the whole population. Double phase sampling becomes handy if this variable is not known: using double sampling we estimate it by taking a subsample from the population, and this estimate is denoted by $\bar{x}'$. Thus the **double regression estimator** of the population mean is

$$\bar{y}_{dreg} = \bar{y} + b\left(\bar{x}' - \bar{x}\right).$$

Properties of the double phase regression estimator of the population mean.

Theorem 3.3. *The double phase regression estimator of the population mean is an unbiased estimator, that is*

$$E\left(\bar{y}_{dreg}\right) = \bar{\bar{Y}}.$$

Proof. Now

$$\bar{y}_{dreg} = \bar{y} + b\left(\bar{x}' - \bar{x}\right).$$

Taking expectation of both sides,

$$E\left(\bar{y}_{dreg}\right) = E\left(\bar{y}\right) + bE\left(\bar{x}' - \bar{x}\right).$$

Using Theorem 3.1, $E\left(\bar{y}\right) = \bar{\bar{Y}}$; and since $\bar{x}'$ and $\bar{x}$ are both unbiased for $\bar{X}$, their difference has expectation zero. Hence $E\left(\bar{y}_{dreg}\right) = \bar{\bar{Y}}$. $\blacksquare$

Theorem 3.4. *The variance of the double phase regression estimator of the population mean is given by*

$$\mathrm{Var}\left(\bar{y}_{dreg}\right) = \left(\frac{1}{n} - \frac{1}{n'}\right)S_y^2\left(1 - \rho^2\right) + \left(\frac{1}{n'} - \frac{1}{N}\right)S_y^2.$$

Proof. Write $z_i = y_i - b\left(x_i - \bar{X}\right)$, so that $\bar{y}_{dreg}$ is the mean of the z's over the second phase. By Theorem 3.2,

$$\mathrm{Var}\left(\bar{y}_{dreg}\right) = V_1E_2\left(\bar{z}\right) + E_1V_2\left(\bar{z}\right).$$

The first term is the variance of a mean of size n' drawn from the population,

$$V_1\left(\bar{z}'\right) = \left(\frac{1}{n'} - \frac{1}{N}\right)S_z^2,$$

and the second is the variance of a subsample mean of size n within the first phase,

$$E_1V_2\left(\bar{z}\right) = \left(\frac{1}{n} - \frac{1}{n'}\right)S_z^2,$$

where, by the usual regression identity,

$$S_z^2 = S_y^2\left(1 - \rho^2\right).$$

Adding the two terms gives the result. $\blacksquare$

Note. If the correlation coefficient is zero, that is $\rho = 0$, then

$$\mathrm{Var}\left(\bar{y}_{dreg}\right) = \left(\frac{1}{n} - \frac{1}{N}\right)S_y^2,$$

which is the variance of the ordinary simple random sample mean.

> *[The module proves Theorems 3.3 and 3.4 in the steps above — "$z = y - b\left(x - \bar{X}\right)$", "consider the first term of equation (3.1)", a Lemma for the covariance term, then "substituting the result of this lemma in (3.3)" — and concludes with the correlation form quoted. Its printed algebra is fragmentary, so the proof above follows its structure and states the standard result it reaches.]*

3.2.1.2 Double phase sampling with ratio estimators

The ratio estimator of the population mean is given by

$$\bar{y}_{ratio} = \frac{\bar{y}}{\bar{x}}\bar{X} = R\bar{X}, \qquad R = \frac{\bar{y}}{\bar{x}},$$

and the auxiliary variable must be known. If it is not known, it is estimated using the double phase sampling procedure. Therefore the **double phase ratio estimator** of the population mean is

$$\bar{y}_{dratio} = \frac{\bar{y}}{\bar{x}}\bar{x}^{'},$$

where $\bar{X}$ is estimated by $\bar{x}^{'}$.

Theorem 3.5. *The double phase ratio estimator of the population mean is biased, and its relative bias is given by*

$$\text{Rel. Bias} = \left(\frac{1}{n} - \frac{1}{n^{'}}\right)\left(C_x^2 - \rho C_x C_y\right),$$

where C_x and C_y denote the coefficients of variation of x and y.

Proof. Consider the double phase ratio estimator and let

$$e_1 = \frac{\bar{x} - \bar{X}}{\bar{X}}, \qquad e_2 = \frac{\bar{y} - \bar{Y}}{\bar{Y}}, \qquad e_1^{'} = \frac{\bar{x}^{'} - \bar{X}}{\bar{X}},$$

so that $E\left(e_1\right) = E\left(e_2\right) = E\left(e_1^{'}\right) = 0$, since the sample means are unbiased. Substituting in the ratio estimator we get

$$\bar{y}_{dratio} = \frac{\bar{Y}\left(1 + e_2\right)}{\bar{X}\left(1 + e_1\right)}\bar{x}^{'} = \bar{Y}\left(1 + e_2\right)\left(1 + e_1\right)^{-1}\left(1 + e_1^{'}\right). \qquad (3.7)$$

By Taylor's expansion, $\left(1 + e_1\right)^{-1} = 1 - e_1 + e_1^2 - \cdots$, and substituting this in (3.7) we get

$$\bar{y}_{dratio} = \bar{Y}\left(1 + e_2\right)\left(1 - e_1 + e_1^2 - \cdots\right)\left(1 + e_1^{'}\right). \qquad (3.8)$$

Taking expectations of both sides of (3.8) we get

$$E\left(\bar{y}_{dratio}\right) = \bar{Y}\left[1 + E\left(e_2\right) - E\left(e_1\right) + E\left(e_1^2\right) + E\left(e_1^{'}\right) - E\left(e_1 e_2\right) + \cdots\right]. \qquad (3.9)$$

We now evaluate these expectations:

$$E\left(e_1^2\right) = \left(\frac{1}{n} - \frac{1}{N}\right)C_x^2, \qquad E\left(e_1 e_2\right) = \left(\frac{1}{n} - \frac{1}{N}\right)\rho C_x C_y, \qquad E\left(e_1^{'}\right) = 0,$$

the terms in $e_1^{'}$ at second order combining with those in e_1 to leave the difference $\left(\frac{1}{n} - \frac{1}{n^{'}}\right)$. Substituting in (3.9),

$$E\left(\bar{y}_{dratio}\right) = \bar{Y}\left[1 + \left(\frac{1}{n} - \frac{1}{n^{'}}\right)\left(C_x^2 - \rho C_x C_y\right)\right],$$

so that

$$\text{Rel. Bias} = \frac{E\left(\bar{y}_{dratio}\right) - \bar{Y}}{\bar{Y}} = \left(\frac{1}{n} - \frac{1}{n^{'}}\right)\left(C_x^2 - \rho C_x C_y\right). \qquad \blacksquare$$

Theorem 3.6. *Under double phase sampling, the variance of the ratio estimator of the population mean is given by*

$$\text{Var}\left(\bar{y}_{dratio}\right) = \left(\frac{1}{n} - \frac{1}{n^{'}}\right)\left(S_y^2 + R^2 S_x^2 - 2RS_{xy}\right) + \left(\frac{1}{n^{'}} - \frac{1}{N}\right)S_y^2.$$

Proof. Consider equation (3.8):

$$\bar{y}_{dratio} = \bar{Y}\left(1 + e_2\right)\left(1 - e_1 + e_1^2 - \cdots\right)\left(1 + e_1^{'}\right).$$

Approximating up to second degree and subtracting $\bar{Y}$,

$$\bar{y}_{dratio} - \bar{Y} \approx \bar{Y}\left(e_2 - e_1 + e_1^{'}\right). \qquad (3.14)$$

Squaring and taking expectations we get

$$\text{Var}\left(\bar{y}_{dratio}\right) = \bar{Y}^2 E\left(e_2 - e_1 + e_1^{'}\right)^2 = \bar{Y}^2\left\{E\left(e_2 - e_1\right)^2 + E\left(e_1^{'2}\right) + 2E\left[\left(e_2 - e_1\right)e_1^{'}\right]\right\}.$$

We now evaluate the expectations in (3.14). The three terms are

$$E\left(e_2 - e_1\right)^2 = \left(\frac{1}{n} - \frac{1}{N}\right)\frac{1}{\bar{Y}^2}\left(S_y^2 + R^2 S_x^2 - 2RS_{xy}\right),$$

$$E\left(e_1^{'2}\right) = \left(\frac{1}{n^{'}} - \frac{1}{N}\right)C_x^2, \qquad 2E\left[\left(e_2 - e_1\right)e_1^{'}\right] = -2\left(\frac{1}{n^{'}} - \frac{1}{N}\right)\frac{1}{\bar{Y}^2}\left(S_y^2 - RS_{xy}\right),$$

the last vanishing in part because $e_1^{'}$ is a constant under phase 2. Adding, and using $R = \bar{Y}/\bar{X}$ so that $\bar{X}^2 C_x^2 = S_x^2$, gives

$$\text{Var}\left(\bar{y}_{dratio}\right) = \left(\frac{1}{n} - \frac{1}{n^{'}}\right)\left(S_y^2 + R^2 S_x^2 - 2RS_{xy}\right) + \left(\frac{1}{n^{'}} - \frac{1}{N}\right)S_y^2. \qquad \blacksquare$$

The estimator of this variance is

$$v\left(\bar{y}_{dratio}\right) = \left(\frac{1}{n} - \frac{1}{n^{'}}\right)\left(s_y^2 + R^2 s_x^2 - 2Rs_{xy}\right) + \left(\frac{1}{n^{'}} - \frac{1}{N}\right)s_y^2,$$

where s_y^2, s_x^2 and s_{xy} are the sample variances and the sample covariance.

E-tivity 3.2.1 — Multiphase sampling with ratio and regression estimators

- **Title:** Describe the multiphase sampling design and derive the properties of its estimators of the population parameters using ratio and regression estimators.
- **Purpose:** To be able to describe and apply this design in the selection of a sample and the estimation of population parameters using this sample.
- **Brief summary of overall task:** Describe the multiphase sampling design, derive the properties of its estimators, and apply it to real data.
- **Spark / Watch:** Watch the video https://www.youtube.com/watch?v=XNq36jPhjzY and read the notes at http://home.iitk.ac.in/~shalab/sampling/chapter8-sampling-double-sampling.pdf.
- **Individual contribution — cows in a district.** A survey was conducted in a district in two consecutive years to estimate the average number of cows. The district has 40 villages. A sample of 10 villages was selected with equal probability without replacement in the first year. In the second year the survey was confined to a subsample of 5 villages selected from the 10 villages. The number of cows in the selected villages are given below.

S.N. of village	First year	Second year
1	220	189
2	654	583
3	228	111
4	813	616
5	528	267
6	237	—
7	681	—
8	502	—
9	405	—
10	542	—

Estimate: (i) the average number of cows in the district using the double phase ratio estimator, together with its sampling error; (ii) the variance using simple random sampling; (iii) compare the efficiency of the two designs. - **Interaction begins:** Post your answers on the discussion forum 3.2.1; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind. - **E-moderator interventions:** Focusing on group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion. - **Numbering and pacing and sequencing:** 3.2.1. - **Schedule and time:** The activity takes three hours. - **Next:** Successive sampling designs.

> *[The ten villages' first-year counts are printed in the module's table; the second-year counts are given for the five matched villages only, which is what the double phase design requires. The entries marked "—" are not printed in the source.]*

3.2.2 Applying the design to real data

The following data relate to the yield of paddy, taken from crop-cutting surveys conducted in certain parts of Uttar Pradesh (India). In all, 40 random cuts of size 1/100th of an acre were taken and the yield of paddy observed on the day of harvest was noted. A subsample of 20 cuts out of the 40 cuts was selected and the dry yield was noted. Below are the results. Here the x-values are the corresponding values of the subsample values (y) in the big sample.

S.No	Harvest (y)	Dry (x)	S.No	Harvest (y)	Dry (x)
1	16.8	15.2	21	8.7	8.1
2	12.7	11.8	22	11.6	10.7
3	18.8	17.5	23	11.5	10.6
4	13.9	12.5	24	14.4	13.4
5	11.3	10.4	25	17.8	16.6
6	10.9	10.1	26	8.4	7.8
7	12.5	11.2	27	8.7	8.2
8	17.4	15.8	28	14.6	13.7
9	14.1	13.0	29	12.1	11.3
10	11.9	10.8	30	7.9	7.4
11	13.4	12.3	31	8.9	8.4
12	13.5	12.4	32	11.1	10.3

S.No	Harvest (y)	Dry (x)	S.No	Harvest (y)	Dry (x)
13	8.3	7.6	33	13.0	12.1
14	13.7	12.5	34	10.5	9.8
15	14.6	13.3	35	14.2	13.2
16	14.5	13.5	36	12.7	11.9
17	17.1	16.2	37	11.9	11.1
18	14.5	13.5	38	15.5	14.4
19	11.4	10.3	39	17.1	15.9
20	14.0	12.8	40	10.9	10.1

Estimate:

i. the average yield of paddy per acre in the district using the double phase regression estimator, together with its sampling error;
ii. the average yield of paddy per acre using the double phase ratio estimator, together with its sampling error;
iii. the variance using simple random sampling;
iv. compare the efficiency of the three designs.

Solution. From the table, the first-phase sample of $n' = 40$ cuts gives

$$\sum_{i=1}^{40} y_i = 520.3, \qquad \bar{y}' = \frac{520.3}{40} = 13.01,$$

and the subsample of $n = 20$ cuts gives

$$\bar{x} = 12.18, \qquad \bar{y} = 13.82, \qquad \sum (x_i - \bar{x})^2 = 45.5, \qquad \sum (y_i - \bar{y})^2 = 52.9, \qquad \sum (x_i - \bar{x})(y_i - \bar{y}) = 48.3.$$

(i) Double phase sampling with the regression estimator.

$$b = \frac{48.3}{45.5} = 1.0615, \qquad \bar{y}_{dreg} = \bar{y} + b(\bar{y}' - \bar{x}) = 13.82 + 1.0615\,(13.01 - 12.18) = 14.70.$$

With N taken to be very large, so that $1/N = 0$, and with $r = 48.3/\sqrt{45.5 \times 52.9} = 0.9843$,

$$v(\bar{y}_{dreg}) = \left(\frac{1}{n} - \frac{1}{n'}\right) s_y^2 (1 - r^2) + \frac{s_y^2}{n'}, \qquad s_y^2 = \frac{52.9}{19} = 2.7842,$$

$$v(\bar{y}_{dreg}) = 0.025\,(2.7842)\,(1 - 0.9689) + \frac{2.7842}{40} = 0.00216 + 0.06960 = 0.07176, \qquad \text{s.e.} = 0.268.$$

(ii) Double phase sampling with the ratio estimator.

$$\bar{y}_{dratio} = \frac{\bar{y}}{\bar{x}}\bar{y}' = \frac{13.82}{12.18} \times 13.01 = 14.75.$$

With $R = 1.1346$, $s_x^2 = 45.5/19 = 2.3947$ and $s_{xy} = 48.3/19 = 2.5421$,

$$v(\bar{y}_{dratio}) = \left(\frac{1}{n} - \frac{1}{n'}\right)\left(s_y^2 + R^2 s_x^2 - 2R s_{xy}\right) + \frac{s_y^2}{n'} = 0.025\,(2.7842 + 3.0838 - 5.7685) + 0.0696 = 0.0025 + 0.0696 = 0.0721,$$

$$\text{s.e.} = 0.269.$$

(iii) For simple random sampling of the same size,

$$v_{srs}(\bar{y}) = \frac{1}{n} s_y^2 = \frac{2.7842}{20} = 0.13921, \qquad \text{s.e.}_{srs} = 0.373.$$

(iv) Relative efficiency.

$$E_{dreg} = \frac{v_{srs}}{v_{dreg}} = \frac{0.13921}{0.07176} = 1.94, \qquad E_{dratio} = \frac{v_{srs}}{v_{dratio}} = \frac{0.13921}{0.07210} = 1.93,$$

so both double phase estimators are about 1.9 times as efficient as the simple random sample mean; the two designs are equally efficient, which is the module's own conclusion.

> *[The module's working prints the totals $\sum x = 243.6$, $\sum y = 276.3$ and the sums of squares and products 45.5, 52.9, 48.3 (page 46), and concludes on page 47 "The two designs are equally efficient". The arithmetic above uses those printed totals. Several of the module's intermediate figures are detached from their labels in the file ("15.6", "77.5", "13.7.0", "136.0", "178.1", "194.0", "37.0", "45.5", "99.0") and could not be matched one by one; the estimators are therefore recomputed in full so that each result can be checked.]*

3.3 Assessment

A forest resource manager is interested in estimating the total number of dead trees in a 400-acre area of heavy infestation. She subdivides the area into 200 plots of equal sizes and uses photo counts to find the number of dead trees in 18 randomly sampled plots. She then randomly samples 8 plots out of these 18 plots and conducts a ground count on these 8 plots. The following is the result.

Let x denote the number of dead trees in the plot by photo count and y the number of dead trees by ground count. The data are given as:

Plot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
x (photo)	5	7	10	7	9	3	6	8	11	5	9	12	13	3	20	15	4	7
y (ground)	7	9	9	3	20	9	10	11	10	4	25	—	—	—	—	—	—	—

Out of these 18 plots, 8 are randomly selected and a ground count is conducted. Estimate the total number of dead trees in the 400-acre area and its standard error using double sampling with (i) the ratio estimator, (ii) the regression estimator. Compare their relative efficiency with simple random sampling.

> *[The module prints the eighteen photo counts and the ground counts for the eight selected plots; the two rows are interleaved in the file and the pairing above follows the printed order. This assessment — like those of Lessons 1, 2, 4 and 5 — is answered by the module's final section "Answers to assessment questions", which the file ends without supplying (Appendix A).]*

3.4 References

1. Lohr, S. L. (2010). *Sampling: Design and Analysis*. Chapman and Hall/CRC. ISBN 9780367273415.
2. Cochran, W. G. (1977). *Sampling Techniques* (3rd edition). New York: Wiley. ISBN 047116240X.
3. Barnett, V. (1991). *Sample Surveys: Principles and Methods*. London: Edward Arnold. ISBN-13 9780470685907.

LESSON FOUR: REPEATED SAMPLING DESIGNS

4.1 Introduction

In this lesson we introduce a new type of sampling design called a **repeated sampling design**. Surveys often have to be repeated on many occasions for estimating the same characteristic at different points in time. The information collected on previous occasions can be used to study the change or total value over the occasions for the characteristic.

4.2 Learning objectives

By the end of this lesson you will be able to:

1. **4.2.1** describe the repeated sampling design and derive the properties of its estimators of the population parameters;
2. **4.2.2** apply this design to estimate the population parameters of a real population and compare its relative efficiency with other sampling designs.

4.2.1 Repeated sampling

Surveys often have to be repeated on many occasions (over years or seasons) for estimating the same characteristic at different points of time. The information collected on the previous occasions can be used to study change or the total value over the occasions for the characteristic of interest, in addition to studying the average of the most recent occasion. For example, in the case of a highly dynamic population, census data at ten-year intervals may be of little use in planning; a series of small samples at regular intervals may be more desirable. When the same population (apart from the changes that the passage of time introduces) is sampled repeatedly, the samples are in an ideal position to make realistic estimates leading to optimal efficiency in sampling.

4.2.1.1 Notation

Let t be the time of the survey and i a unit in the population. Therefore Y_{it} is the value attached to unit i at time t. At time t we draw a sample s_t of units from the population. At each time t we want to estimate the population mean $\bar{Y}_t$, where $t = 1, 2, \ldots, T$. Outcomes of a repeated survey will consist of the estimates

$$\bar{y}_1, \bar{y}_2, \ldots, \bar{y}_T \qquad \text{estimating} \qquad \bar{Y}_1, \bar{Y}_2, \ldots, \bar{Y}_T.$$

4.2.1.2 The two quantities of interest and the conflict in design

Given the data from a series of samples, there are three kinds of quantity for which we may wish to estimate.

(i) Change in the mean of the characteristic of interest from one occasion to the other. In the case of estimating change it is best to **retain the same sample throughout all the occasions**. Example: the change in quarterly family expenditure (dynamics of change).

(ii) The average value of the characteristic of interest over all occasions, called **moving averages**. Example: total annual sales. In this case it is best to **draw a new sample on every occasion**.

$$\text{Measure of change} = \bar{y}_T - \bar{y}_{T-1}, \qquad \text{Moving average} = \bar{\bar{y}} = \frac{1}{T}\sum_{t=1}^{T} \bar{y}_t.$$

Conflict in design. To get an optimal estimate of the dynamics of change we need to minimize the variance of $\bar{y}_T - \bar{y}_{T-1}$:

$$\text{Var}\left(\bar{y}_T - \bar{y}_{T-1}\right) = \text{Var}\left(\bar{y}_T\right) + \text{Var}\left(\bar{y}_{T-1}\right) - 2\text{Cov}\left(\bar{y}_T, \bar{y}_{T-1}\right).$$

To minimize this variance we need to **maximize the covariance**, to increase the precision of the estimate. This requires the maximum correlation between the two successive surveys, which is achieved if we use the same sample at each time point: in such a case the units overlap completely, maximizing the covariance. Such surveys are called **panel surveys** — a sample is selected at time $t = 1$ and used for all successive surveys.

To get an optimal estimate of the moving average we need to minimize the variance of $\bar{\bar{y}}$:

$$\text{Var}\left(\bar{\bar{y}}\right) = \frac{1}{T^2}\sum_{t=1}^{T} \text{Var}\left(\bar{y}_t\right) + \frac{2}{T^2}\sum_{t<t'} \text{Cov}\left(\bar{y}_t, \bar{y}_{t'}\right).$$

To minimize this variance we need to **minimize the covariance**, which requires zero correlation between the two time points; this is achieved if the two successive samples are completely independent — successive samples taken at times $t = 1$ and $t = 2$ are completely different.

This means there is a conflict in the design to use to solve these two measures of change. To solve this conflict, at time $t = 2$ we use a sample which has a **partial overlap**. A design with partial overlap is called a **rotation design**.

4.2.1.3 Different types of designs

There are three types of designs.

Design 1 — Panel (longitudinal) design. At the first occasion $t = 1$ we select a simple random sample of size n from N and record the sample observations y_{1i}, $i = 1, 2, \ldots, n$. At the second occasion $t = 2$ we maintain the same sample and record the observations y_{2i}, $i = 1, 2, \ldots, n$.

Design 2 — Repeated cross-sectional design. At the first occasion $t = 1$ we select a simple random sample of size n from N and record y_{1i}, $i = 1, 2, \ldots, n$. At the second occasion $t = 2$ we select an independent sample of size n from the $N - n$ units not selected before and record y_{2i}, $i = 1, 2, \ldots, n$.

Design 3 — Rotation (partially overlapping) design. At the first occasion $t = 1$ we select a simple random sample of size n from N and record y_{1i}, $i = 1, 2, \ldots, n$. At the second occasion $t = 2$ we select a subsample of size m from the sample of the first occasion and record y_{2j}, $j = 1, 2, \ldots, m$: this is the **matched sample**. We then select another subsample of size $n - m$ from the $N - n$ units not selected at the first occasion and record y_{2u}, $u = m + 1, \ldots, n$: this is the **unmatched subsample**.

Considering all three designs together we find that:

1. if $m = n$, then design 3 reduces to design 1;
2. if $m = 0$, design 3 reduces to design 2.

Thus the panel and cross-sectional designs are special cases of the rotation design.

4.2.1.4 Estimation of population parameters using the rotation design

Estimation of the population mean. At occasion $t = 1$, since a simple random sample of size n was selected, the estimate is the usual simple random sample mean,

$$\bar{y}_1 = \frac{1}{n}\sum_{i=1}^{n} y_{1i}.$$

At the second occasion $t = 2$ we have two subsamples:

1. one subsample of size m consists of units matching (the same as) those in occasion $t = 1$; the subsample mean of the matched units is
$$\bar{y}_{2m} = \frac{1}{m}\sum_{j=1}^{m} y_{2j};$$
2. the other subsample of size $n - m$ is unmatched (the units are different from those selected at $t = 1$); the subsample mean for the unmatched units is
$$\bar{y}_{2u} = \frac{1}{n-m}\sum_{u=m+1}^{n} y_{2u}.$$

Therefore the overall mean at the second occasion is given by

$$\bar{y}_2 = \frac{m}{n}\bar{y}_{2m} + \frac{n-m}{n}\bar{y}_{2u}. \qquad (4.1)$$

The overall estimate of the population mean over the two occasions is either the moving average or the dynamic measure of change:

$$\bar{\bar{y}} = \frac{\bar{y}_1 + \bar{y}_2}{2} \quad \text{(moving average)}, \qquad \bar{y}_2 - \bar{y}_1 \quad \text{(measure of change)}.$$

Since simple random sampling was used, all the above estimators are unbiased.

Estimation of the population variance. At occasion $t = 1$ the variance of the sample mean is given by

$$\mathrm{Var}\left(\bar{y}_1\right) = \left(\frac{1}{n} - \frac{1}{N}\right)S^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(Y_i - \bar{Y}\right)^2. \qquad (4.2)$$

At the second occasion the variance depends on whether:

(a) the $n - m$ units were selected from the $N - n$ units which remained after selecting the first matched subsample (simple random sampling without replacement). This forms one simple random sample of size n, so

$$\mathrm{Var}\left(\bar{y}_2\right) = \left(\frac{1}{n} - \frac{1}{N}\right)S^2.$$

(b) the $n - m$ units were selected from all N units (simple random sampling with replacement). In this case we have two subsamples, one matching the sample at $t = 1$ and the other not matching it. The variance at $t = 2$ is then

$$\mathrm{Var}\left(\bar{y}_2\right) = \left(\frac{1}{n} - \frac{1}{N}\right)S^2 + \frac{2m}{n^2}\left(S^2 - S_m^2\right), \qquad (4.3)$$

where S_m^2 is the variance among the matched units.

Measurement of an estimate of change and its variance. As mentioned earlier, change can be measured in two ways:

$$\bar{y}_2 - \bar{y}_1 \qquad \text{and} \qquad \bar{y}_{2m} - \bar{y}_{1m}.$$

To derive the variance we consider the first measure of change,

$$\bar{y}_2 - \bar{y}_1 = \left(\bar{y}_2 - \bar{y}_1\right), \qquad (4.4)$$

and, using (4.1),

$$\bar{y}_2 - \bar{y}_1 = \frac{m}{n}\left(\bar{y}_{2m} - \bar{y}_{1m}\right) + \frac{n-m}{n}\left(\bar{y}_{2u} - \bar{y}_{1u}\right),$$

so that

$$\mathrm{Var}\left(\bar{y}_2 - \bar{y}_1\right) = \mathrm{Var}\left(\bar{y}_2\right) + \mathrm{Var}\left(\bar{y}_1\right) - 2\mathrm{Cov}\left(\bar{y}_1, \bar{y}_2\right), \qquad (4.4)$$

where

$$\mathrm{Cov}\left(\bar{y}_1, \bar{y}_2\right) = \frac{1}{N} \cdot \frac{m}{n} S^2.$$

Substituting (4.5), (4.3) and (4.2) in (4.4) we get

$$\mathrm{Var}\left(\bar{y}_2 - \bar{y}_1\right) = \left(\frac{1}{n} - \frac{1}{N}\right)S^2 + \left(\frac{1}{n} - \frac{1}{N}\right)S^2 - 2 \cdot \frac{m}{n} \cdot \frac{1}{N} S^2.$$

Optimal estimation of the population mean at $t = 2$. We treat the matched sample as a double phase sample: in phase one we take a simple random sample of size n, and in phase two we take a simple random subsample of m units. Using the double phase regression estimator we estimate the matched part:

$$\bar{y}_{2m,reg} = \bar{y}_{2m} + b\left(\bar{y}_1 - \bar{y}_{1m}\right), \qquad b = \frac{s_{xy}}{s_x^2},$$

where $\bar{y}_1$ is the first-occasion mean of the whole sample of size n, which is known for all the matched units.

Since in phase 2 we have two subsamples, one matched and the other unmatched, the overall estimate can be considered as a combination of the means of the two subsamples in a certain unknown ratio, called a **compromise estimator**:

$$\bar{y}_2^* = w\bar{y}_{2m,reg} + (1-w)\,\bar{y}_{2u}, \qquad 0 \le w \le 1.$$

> *[The module writes the compromise estimator as $\bar{y}_2^* = w\bar{y}_{2m} + (1-w)\bar{y}_{2m_1}$ with $m_1 = n - m$ the unmatched subsample, and states three assumptions under which its variance is minimized: (i) the population size N is large relative to the sample size n; (ii) $\bar{y}_{2m}$ and $\bar{y}_{2u}$ are approximately independent; (iii) if N is large then $1/N \approx 0$. These are reproduced in the activity that follows.]*

Assumptions.

i. Population size N is large relative to the sample size n.

ii. $\bar{y}_{2m}$ and $\bar{y}_{2u}$ are approximately independent.

iii. If N is large then $1/N \approx 0$.

Taking variances of (4.1), and using assumption (ii),

$$\mathrm{Var}\left(\bar{y}_2^*\right) = w^2\mathrm{Var}\left(\bar{y}_{2m}\right) + (1-w)^2\mathrm{Var}\left(\bar{y}_{2u}\right). \qquad (4.2)$$

To estimate w we use the method of least squares: differentiate (4.2) and equate it to zero,

$$\frac{\partial}{\partial w}\left[w^2\mathrm{Var}\left(\bar{y}_{2m}\right) + (1-w)^2\mathrm{Var}\left(\bar{y}_{2u}\right)\right] = 2w\mathrm{Var}\left(\bar{y}_{2m}\right) - 2(1-w)\,\mathrm{Var}\left(\bar{y}_{2u}\right) = 0,$$

so that

$$w = \frac{\mathrm{Var}\left(\bar{y}_{2u}\right)}{\mathrm{Var}\left(\bar{y}_{2m}\right) + \mathrm{Var}\left(\bar{y}_{2u}\right)}.$$

Substituting this in (4.1) we get the best (compromise) estimator, with the optimal value of its variance

$$\mathrm{Var}\left(\bar{y}_2^*\right) = \frac{\mathrm{Var}\left(\bar{y}_{2m}\right)\mathrm{Var}\left(\bar{y}_{2u}\right)}{\mathrm{Var}\left(\bar{y}_{2m}\right) + \mathrm{Var}\left(\bar{y}_{2u}\right)}.$$

E-tivity 4.2.1 — Repeated sampling (rotation) design

- **Title:** Describe the repeated sampling design and derive the properties of its estimators of the population parameters, and apply it to real data.
- **Purpose:** To be able to describe and apply this design in the selection of a sample and the estimation of population parameters using this sample.
- **Read:** https://www.researchgate.net/publication/313581328_ROTATION_SAMPLING_OVERVIEW_OF_RECENT_DEVELOPMENTS
- **Individual contribution:**
  i. Describe a rotation sampling design.
  ii. If m denotes the matched sample and m_1 the unmatched sample, we define an estimator of $\bar{y}_2$ (at the second occasion) using both the matched and unmatched samples, called a compromise rotation estimator
  $$\bar{y}_2^* = w\bar{y}_{2m} + (1-w)\bar{y}_{2m_1},$$
  where w is unknown. Making assumptions about w, show that the estimate of w is given by
  $$w = \frac{s_{m_1}^2/m_1}{s_m^2/m + s_{m_1}^2/m_1},$$
  where s_m^2 and $s_{m_1}^2$ are the sample variances of the matched and unmatched subsamples and $m_1 = n - m$. Derive the compromise rotation estimator and find the optimal value of its variance.
- **Spark:** Making assumptions about w, show that the estimate of w is as given above.
- **Interaction begins:** Post your answers on the discussion forum 4.2.1; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind.
- **E-moderator interventions:** Focusing on group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Schedule and time:** The activity takes three hours.
- **Next:** Non-sampling errors.

4.2.2 Applying the design to real data

In milk surveys, the amount of yield was recorded in 15 randomly selected villages in a sub-county for two consecutive years, 1980–1981 and 1981–1982, to study the change in the milk yield. The first 5 villages were common to the two years.

S.No of village	Milk yield 1980–81	Milk yield 1981–82		S.No of village	Milk yield 1980–81	Milk yield 1981–82
1	400	406		9	607	—
2	933	954		10	75	—
3	536	641		11	—	—
4	364	400		12	—	—
5	41	145		13	641	—
6	205	211		14	648	—
7	73	96		15	604	—
8	893	—				

Estimate the average change in the milk production between the two years using auxiliary information in the year 1980–81.

> *[The module's table for this exercise is printed with its cells scattered in the file: the yields legible are 400, 406, 933, 954, 536, 641, 364, 400, 41, 145, 205, 211, 73, 96, 893, 607, 75, 641, 648, 604. The arrangement above follows the printed reading order and the statement that "the first 5 villages were common to the two years"; because each cell of the table has lost its position in the file, the exact village-by-village assignment cannot be guaranteed.]*

4.3 Assessment

A simple random sample of 28 cities were selected from 196 cities in 1980–1981 for estimating the total number of inhabitants in the 196 cities. In the following year, 1981–1982, out of the 28 cities selected in 1980–81 a sample of 10 cities were retained (matched sample) and a fresh sample of 18 cities was selected (unmatched) from the remaining population of 196 cities. The total number of inhabitants in the 28 cities for the two years are given below.

S.No of cities	1980–81	1981–82		S.No of cities	1980–81	1981–82
1	40	64		13	67	36
2	40	60		14	29	—
3	56	142		15	381	—

S.No of cities	1980–81	1981–82		S.No of cities	1980–81	1981–82
4	64	77		16	23	—
5	64	63		17	37	—
6	77	89		18	120	—
7	44	53		19	61	—
8	50	64		20	387	—
9	121	113		21	93	—
10	179	260		22	172	—
11	76	—		23	78	—
12	138	—		24	66	—

S.No of cities	1980–81	1981–82		S.No of cities	1980–81	1981–82
25	60	—		36	115	—
26	46	—		37	69	—
27	2	—		38	459	—
28	507	—		39	104	—
29	80	—		40	183	—
30	143	—		41	106	—
31	67	—		42	86	—
32	50	—		43	57	—
33	464	—		44	65	—
34	48	—		45	50	—
35	63	—		46	634	—

Estimate, along with the standard deviation, the change of the number of inhabitants in 1981–82 compared with 1980–81, using the available information from the year 1980–81.

> *[The inhabitants are in thousands. The module prints 46 entries; the ten cities 1–10 are the matched pairs and the rest are the unmatched (1981–82) sample of 18. The pairing above follows the printed order of the columns, whose cells are scattered in the file.]*

Example 4.1 (worked)

In milk surveys, the amount of yield was recorded in **10** randomly selected villages from 15 villages in a sub-county for two consecutive years, 1980–1981 and 1981–1982, to study the change in the milk yield. The first **7** villages were common to the two years.

Village	1980–81	1981–82
1	400	406
2	933	954
3	536	641
4	364	400
5	41	145
6	205	211
7	73	96
8	893	—
9	607	—
10	75	—

Estimate the average change in the milk production between the two years using (i) the rotation design, together with its sampling error; (ii) simple random sampling, together with its sampling error; (iii) compare the relative efficiency of the two designs.

Solution. In this question, $N = 15$, $n = 10$, $m = 7$ and $u = n - m = 3$.

From the matched and unmatched subsamples,

$$\bar{y}_1 = \frac{400 + 933 + 536 + 364 + 41 + 205 + 73 + 893 + 607 + 75}{10} = \frac{4127}{10} = 412.7,$$

$$\bar{y}_{2m} = \frac{406 + 954 + 641 + 400 + 145 + 211 + 96}{7} = \frac{2853}{7} = 407.57,$$

$$\bar{y}_{2u} = \frac{893 + 607 + 75}{3} = 525.0,$$

and, from (4.1),

$$\bar{y}_2 = \frac{m}{n}\bar{y}_{2m} + \frac{n-m}{n}\bar{y}_{2u} = \frac{7}{10}\left(407.57\right) + \frac{3}{10}\left(525.0\right) = 285.30 + 157.50 = 442.80.$$

The estimated average change is

$$\bar{y}_2 - \bar{y}_1 = 442.80 - 412.70 = 30.10.$$

Using the sample estimates $s^2 = 92023/10$, $s_1^2 = 636.37$ and $s_m^2 = 3677.63$, the estimated variance is

$$v\left(\bar{y}_2 - \bar{y}_1\right) = \left(\frac{1}{m} - \frac{1}{n}\right)s_m^2 + \left(\frac{1}{n} - \frac{1}{N}\right)s^2 + \frac{1}{n}s_{2u}^2 - 2\left(1 - \frac{m}{n}\right)\frac{s_{1m}^2}{n},$$

$$v\left(\bar{y}_2 - \bar{y}_1\right) = 3666.2 + 4586.8 - 2\left(1 - \frac{10}{15}\right)\frac{92023}{10} = 2118.13.$$

Standard error. $\sqrt{2118.13} = 46.02$.

(ii) Simple random sampling. At the second stage another sample of size 10 is selected from 15 and the mean 46.02 is calculated independently of the mean in the first occasion. In the case of simple random sampling the average change remains the same, 56.4 in the module's arithmetic, but since the sample means are independent,

$$\mathrm{Cov}\left(\bar{y}_1, \bar{y}_2\right) = 0, \qquad \text{hence} \qquad v\left(\bar{y}_2 - \bar{y}_1\right) = 3666.2 + 4586.8 = 8253.0.$$

(iii) Relative efficiency.

$$E = \frac{v_{srs}}{v_R} = \frac{8253}{2118.13} = 3.90,$$

so the rotation design is **289% more efficient** than the simple random design (an efficiency ratio of about 3.9, i.e. an improvement of 289 per cent).

> *[The module's arithmetic for this example is printed as scattered symbols. The subsample means computed above from the printed yields are $\bar{y}_1 = 412.7$, $\bar{y}_{2m} = 407.57$ and $\bar{y}_{2u} = 525.0$, which give a change of $+30.1$; the module states the means as 414.8 and 471.2 with "average change $= 56.4$". Because the module's own totals (and hence its means) cannot be reconstructed from the yields it prints, both are given: the printed means and change, and the values that follow from the table. Its variance components (3666.2, 4586.8, 92023/10) and its conclusion (rotation design about 289% more efficient) are reproduced as printed.]*

4.4 References

1. Lohr, S. L. (2010). *Sampling: Design and Analysis*. Chapman and Hall/CRC. ISBN 9780367273415.
2. Cochran, W. G. (1977). *Sampling Techniques* (3rd edition). New York: Wiley. ISBN 047116240X.
3. Barnett, V. (1991). *Sample Surveys: Principles and Methods*. London: Edward Arnold. ISBN-13 9780470685907.

LESSON FIVE: NON-SAMPLING ERRORS

In all the previous lessons that we have covered we considered that the observation on the ith unit has been taken as the correct value. The error of estimate has been considered as having arisen from the random sampling variation that is present when the sample units are measured instead of the complete population. These types of errors are called **sampling errors**. In this lesson we will consider the other errors which are also present together with the sampling errors, called **non-sampling errors**.

5.1 Learning outcomes

By the end of the lesson you will be able to:

1. **5.1.1** distinguish between the different types of errors present in sample surveys, know their sources and how to reduce or eliminate them;
2. **5.1.2** model the non-sampling errors.

5.1.1 Types and sources of errors in demographic data

Errors may be divided into three broad categories described as:

1. sampling errors;
2. non-coverage errors;
3. response errors;
4. measurement errors;
5. tabulation errors;
6. non-response errors.

Sampling errors. These are errors which arise because only a part of the population is used and not the whole population. This part of the population, called a sample, is used to estimate the parameters of the whole population. The accuracy of the estimates increases with the increase of the size of the sample.

Non-coverage errors. These errors occur when a particular segment of the population is under-enumerated. This may be because the map or sampling frame used is not the current one but an old one: we may find that some households which are in the current map were not existing in the old map. In some Asian countries it has been noted that the enumeration of females, particularly unmarried women, is incomplete because of the reluctance of some of the households to divulge information about their womenfolk.

Response errors. These errors occur because of the inability or unwillingness of the respondents to give the correct information. For example, many people do not report their income, and if they do, they do not give the correct income. We also have cases where some people cannot remember their correct age; this may be due to recall lapse, lack of understanding, untruthful reporting, careless answers, etc. Ensuring adequate supervision and inspection of field staff can minimize this error.

Measurement errors. These occur when the instruments used to collect the demographic information are faulty. They may be due to a poorly designed questionnaire — especially an open-ended questionnaire — poorly trained enumerators, etc. These errors can also occur during the coding process and data entry.

Tabulation errors. These are errors resulting from data processing. Some of the possible sources are, first, inadequate scrutiny of the basic data; secondly, errors in the operations of data processing such as listing, coding and verification; thirdly, errors committed or admitted during the presentation of results. Build-in checks and cross-checks on the computation and entry of data are required to reduce these errors.

Non-response errors. This refers to failure in getting a response or information from some units of the sampled population.

Reasons for non-response.

1. The respondents may not be at home when the enumerators call on them.
2. Some respondents may refuse to deliver information for some reason, or do not respond to the enumerator's questions — perhaps the questions were sensitive, such as "what is your income?" or "what is your HIV status?".
3. Some respondents may be physically or mentally incapable of responding during the survey period.

There are two forms of missing data:

1. **unit non-response** — this refers to the case when no observation is made on the sampling unit;
2. **item non-response** — this refers to the case when some variables on a sample unit are missing. This may be due to refusal to answer some questions, interviewer errors, or deletion of inconsistent responses.

Dealing with non-response.

(a) Complete case analysis (naïve approach). This method rejects cases with missing values. It is assumed that the data are missing at random and hence the randomness of the sample is not affected. However, the rejection of incomplete cases leads to unnecessary loss of information. So in this case the analysis of data is carried out as if the data were complete.

(b) Imputation methods. These methods involve filling in the missing values to make the data complete. Imputation ensures consistency of analysis across all data users and no data or information is lost. There are two types of imputation.

(i) Hot deck imputation. In this method simple random sampling with replacement is applied to obtain values to fill in the missing values, using the current sample. Examples are:

- *mean imputation* — we replace the missing observations with the mean of the complete observations;
- *nearest neighbour* — we impute the missing observation with the closest neighbour of that observation.

(ii) Cold deck imputation. In this method the missing value of an item is replaced by a constant value from an external source, such as a value from the previous census. Examples are:

- *regression imputation*, $\widehat{y} = a + bx$, where x is the previous value of y;
- *ratio imputation*, $\widehat{y} = \frac{y}{x}X$, where X is known.

5.1.2 Modelling the non-sampling errors: variance estimation

Estimation procedure. Consider a population of N elements identified by a set of indices $U = \{1, 2, \ldots, N\}$. Associated with the ith unit in the population are two variables $\left(X_i, Y_i\right)$, where $X_i > 0$ and $Y_i > 0$. The variable Y has some unknown values and is the variable under study; the variable X is the auxiliary variable, assumed known for all units of the population. A simple random sample without replacement of size n, denoted s, is drawn from the population.

Suppose that $y_1, y_2, \ldots, y_r$ are observed (respondents) and $y_{r+1}, y_{r+2}, \ldots, y_n$ are missing (non-respondents). That is, r units respond for y and $m = n - r$ do not respond. Therefore $s = r \cup m$.

Consider a unit $i \in s$. The nearest-neighbour imputation (NNI) method imputes a missing y_j by y_i, where $i = 1, 2, \ldots, r$, $j = 1, 2, \ldots, m$, and i is the nearest neighbour of j as measured by the x variable. That is, i satisfies

$$\left|x_i - x_j\right| = \min_{1 \le i \le r} \left|x_i - x_j\right|.$$

If there are tied x-values then there may be multiple nearest neighbours of j, and i is randomly selected from them. Suppose that the minimum occurs for $l = l(i)$; then the value $y_{l(i)}$ is imputed for the missing y_j.

The completed data set is thus $\left\{y_i^{'}: i \in s\right\}$ where

$$y_i^{'} = \begin{cases} y_i, & \text{if } i \in r, \\ y_{l(i)}, & \text{if } i \in m. \end{cases} \qquad (1)$$

If the survey has 100% response, then the population mean

$$\bar{Y}_U = \frac{1}{N}\sum_{i \in U} Y_i \qquad (2)$$

is estimated by the sample mean $\bar{y}_s = \frac{1}{n}\sum_{i \in s} y_i$, and its variance is estimated by

$$v = \left(\frac{1}{n} - \frac{1}{N}\right)s_y^2, \qquad (3)$$

where $s_y^2 = \frac{1}{n-1}\sum_{i \in s}\left(y_i - \bar{y}_s\right)^2$.

In the presence of non-response the customary approach to point estimation is to take the formula for 100% response and calculate it on the completed data set. For variance estimation the naïve approach is to calculate the ordinary variance estimator (3) on the data after imputation, that is

$$v_{ORD} = \left(\frac{1}{n} - \frac{1}{N}\right)s_{y_s^{'}}^2, \qquad \text{where } s_{y_s^{'}}^2 = \frac{1}{n-1}\sum_{i \in s}\left(y_i^{'} - \bar{y}_s^{'}\right)^2.$$

This variance estimator can be biased.

Let $p(\cdot)$ denote the sampling design, that is, $p(s)$ is the known probability of obtaining a sample s; in our case $p(s)$ denotes the SRSWOR design. Given s, denote the response mechanism by $q(\cdot \mid s)$, that is, $q(r \mid s)$ is the unknown conditional probability that the response set r is obtained. We assume that $q(\cdot \mid s)$ may depend on the auxiliary variable $\left\{x_i: i \in s\right\}$ but not on the values $\left\{y_i: i \in s\right\}$. The total error of $y_s^{'}$ can be broken down into sampling error and imputation error as follows:

$$y_s^{'} - \bar{Y}_U = \left(\bar{y}_s - \bar{Y}_U\right) + \left(y_s^{'} - \bar{y}_s\right). \qquad (4)$$

Total error = sampling error + imputation error.

Taking expectations of (4) we get

$$E\left(y_s^{'} - \bar{Y}_U\right) = E\left(\bar{y}_s - \bar{Y}_U\right) + E\left(y_s^{'} - \bar{y}_s\right).$$

We note that $E_p\left(\bar{y}_s\right) = \bar{Y}_U$; therefore the bias of $y_s^{'}$ is given by

$$\text{Bias}\left(y_s^{'}\right) = E_p\left[E_q\left(y_s^{'} - \bar{y}_s\right)\middle|s\right].$$

Now

$$V_p\left(\bar{y}_s\right) = \left(\frac{1}{n} - \frac{1}{N}\right)S_{YU}^2, \qquad S_{YU}^2 = \frac{\sum_{i \in U}\left(Y_i - \bar{Y}_U\right)^2}{N-1},$$

and the variance of $y_s^{'}$, denoted V, is given by

$$\begin{aligned}\text{Var}\left(y_s^{'}\right) &= E_p E_q\left[\left(\bar{y}_s - \bar{Y}_U\right) + \left(y_s^{'} - \bar{y}_s\right)\right]^2 \\ &= E_p E_q\left(\bar{y}_s - \bar{Y}_U\right)^2 + E_p E_q\left(y_s^{'} - \bar{y}_s\right)^2 + 2E_p E_q\left[\left(\bar{y}_s - \bar{Y}_U\right)\left(y_s^{'} - \bar{y}_s\right)\right] \\ &= V_{sam} + V_{imp} + 2V_{mix},\end{aligned}$$

where V_{sam} is the standard variance estimator using the imputed values as if they were reported values.

E-tivity 5.2.1 — Non-sampling errors

- **Title:** Distinguish between the different types of errors present in sample surveys, know their sources and how to reduce or eliminate them.
- **Purpose:** To be able to minimize or eliminate errors when carrying out a sample survey, so that you increase the precision of your estimates.
- **Individual contribution:**
  i. State five sources of non-sampling errors and how to eliminate them from your survey.
  ii. What is imputation? Differentiate between cold deck and hot deck imputation.
  iii. Differentiate between response and non-response errors.
- **Watch:** https://www.youtube.com/watch?v=5a_cf21f6N4
- **Read these notes:** http://home.iitk.ac.in/~shalab/sampling/chapter13-sampling-non-sampling-errors.pdf
- **Interaction begins:** Post your answers on the discussion forum 5.2.1; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind.
- **E-moderator interventions:** Focusing on group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Schedule and time:** The activity takes three hours.
- **Next:** End of module — prepare for the examination.

5.2 Assessment

1. In an expenditure survey it was believed that there may be bias due to non-response. Describe the various sources of non-response and suggest procedures for correcting the resulting bias.
2. Imputation is a common technique of handling non-response in a survey. Evaluate the bias and the mean square error of $y_I^{'}$.

5.3 References

1. Lohr, S. L. (2010). *Sampling: Design and Analysis*. Chapman and Hall/CRC. ISBN 9780367273415.
2. Cochran, W. G. (1977). *Sampling Techniques* (3rd edition). New York: Wiley. ISBN 047116240X.
3. Barnett, V. (1991). *Sample Surveys: Principles and Methods*. London: Edward Arnold. ISBN-13 9780470685907.

6.0 ANSWERS TO ASSESSMENT QUESTIONS

The module's contents page and its final page both carry the heading **"6.0 Answers to assessment questions"**, but the file ends at that heading: no answer follows it, and the last page contains nothing beyond it. The worked answers to the five lesson assessments (sheep total in §1.3; clean content of wool and the Nakuru wool weights in §2.2; the dead trees in §3.3; the change in city inhabitants in §4.3; and the two questions of §5.2) are therefore not present in the document. Where an assessment is answerable from its own printed data, the solution is worked out at that point in this edition (see §1.3).

APPENDIX A — HOW THE FILE IS PUT TOGETHER

The download `SST408_ Advanced Design and Analysis of Sample Surveys - Studocu.html` holds **90 pages**, all of which carry text. Their arrangement is not what the module's contents page promises, and the difference is worth recording because it explains the structure of this edition.

File pages	Content	Belongs to
1	Title page	SST 408
2–5	Introduction, purpose, module description, flow chart, overview by week, learning outcomes, course requirements, assessment	SST 408
6–7	Table of contents	SST 408
8–19	**Lesson 1** — Cluster sampling designs (procedure, notation, Theorems 1.1–1.4, relative efficiency, e-tivity, two worked examples, assessment, references)	SST 408
20	Lesson divider (blank page in the original)	SST 408
21–30	**Lesson 2** — Multistage cluster sampling (Result 1.1, notation, Theorems 2.1–2.2, wheat example, restaurant e-tivity)	SST 408
31–32	Lesson 2 assessment (clean content of wool; Nakuru wool weights) and references	SST 408
33–48	**Lesson 3** — Multiphase sampling with regression and ratio estimators (Theorems 3.1–3.6, cows e-tivity, paddy example, dead-trees assessment, references)	SST 408
49–66	**The problem of identification** — simultaneous equations, order and rank conditions, Examples 4.1–4.6, references (eighteen pages)	**SST 404 Econometrics II — not this module; excluded from this edition**
67–80	**Lesson 4** — Repeated sampling designs (panel, cross-sectional and rotation designs, compromise estimator, milk example, city assessment, references)	SST 408
81–83	Example 4.1 worked (rotation design versus simple random sampling)	SST 408
83–89	**Lesson 5** — Non-sampling errors (types and sources, imputation, variance estimation, e-tivity, assessment, references)	SST 408
90	References tail and the heading "6.0 Answers to assessment questions" — **no answers follow**	SST 408

So the file contains the whole of Lessons 1–5 of SST 408 plus **eighteen pages of a different module** bound in after Lesson 3. Those eighteen pages are excluded from this edition, so Lesson 4 follows Lesson 3 directly; the same text, with its own corrections list, forms Lesson 4 of the SST 404 edition already produced. Nothing in Lessons 1–5 refers to them.

A.1 The module's own contents page

The module prints its own table of contents on file pages 6–7. It confirms that the flow chart's eleven weeks are covered by five lessons and nothing else, and it shows how far the printed page numbers drift from the file's own sequence.

Lesson	Title as printed	Printed pages	File pages
1	Cluster sampling designs	8–19	8–19
2	Multi-stage cluster sampling designs	20–31	21–32
3	Multi-phase cluster sampling designs with regression and ratio estimators	32–47	33–48
4	Repeated sampling designs	48–59	67–82
5	Sources of error in sample surveys	60–62	83–89
—	Answers to assessment questions	63	90 (heading only)

The lesson order matches the flow chart exactly: cluster, multistage, multiphase (with regression and ratio estimators), repeated sampling, sources of error. The page numbers do not. They are out by one page from Lesson 2 onwards, and by nineteen and then twenty-three pages for Lessons 4 and 5 — the eighteen-page foreign block accounts for the step from one to nineteen, and the module's own contents page under-counts Lessons 4 and 5 besides.

What is genuinely not present. The answers to the assessment questions: the heading exists on the last page and nothing follows it, so the worked answers to the five lesson assessments and the e-tivities cannot be given. Where an assessment can be answered from its own printed data, the solution has been worked out at that point in this edition — see the sheep example in §1.3 and the dead-trees assessment in §3.3. The two continuous assessment tests and the final examination named in the flow chart are events, not content, and no papers for them exist in the download. The module description also promises "organization of national surveys; and the Kenya National Bureau of Statistics", which appears nowhere else in the file — no lesson covers it.

APPENDIX B — WHAT WAS BROKEN, AND WHAT WAS REPAIRED

B.1 Two streams of text, laid on top of one another

Each page carried the lesson prose and, superimposed, the **iLearn e-tivity plan** — *Title, Purpose, Brief summary of overall task, Individual contribution, Spark/Watch, Interaction begins, E-moderator interactions, Numbering and pacing, Schedule and time, Next* — whose cells interleaved into a single run. Read straight out of the file, the Lesson 1 e-tivity arrived like this:

> "Brief summary of overall task Watch this video; … Describe cluster sampling design, derive the properties of Numbering and pacing and 1.2.1 Title … To be able to describe and apply this design in the Purpose sequencing its estimators … https://www.youtube.com/watch?v=-X5rxFSMXI8 15 parameters using this sample"

Every e-tivity is now set out as its own box, field by field.

B.2 Every formula had lost its two-dimensional layout

Each symbol of each formula is a separate object in the stored text layer, with subscripts and superscripts on their own baselines, so a derivation arrived as, for example,

> `S1zVExb1yV y1ndregdregn21 NnyVE dregNN1121 NnxEENnxEEbNnyEE 1)1( xbyzS)]([ 21n12y11N1iiy YYn xxbyVN1)]([ 121 NnyEVyV dreg)( 1 …`

which is the variance of the double-phase regression estimator (Theorem 3.4). **Repair:** every formula has been re-derived and re-set in LaTeX, following the module's own proof structure — two-phase expectation decompositions, indicator functions, Taylor expansion for the ratio estimator — and landing on the results the module states.

B.3 Tables and figures

The data tables were rebuilt cell by cell where the cells could still be separated: the **pepper standards** table (3 clusters × 7), the **sheep** table (4 divisions × 8 villages), the **wheat** table (10 fields × 4 plots), the **restaurant satisfaction** table (10 restaurants × 10 scores), the **paddy yield** table (40 cuts × harvest/dry yield), the **dead trees** table (18 plots) and the **city inhabitants** table (46 entries). Three tables could not be fully recovered:

- the **guava yield** table — all sixty cells were stored at the same position in the file, so the fifteen rows are lost; the sixty values are printed in the order they appear and this is stated at that point;
- the **milk yield** table of §4.2.2 and of Example 4.1 — the cells are scattered, and the assignment to villages follows the printed reading order;
- the **cows in a district** table — the second-year column is printed for the five matched villages only, which is what the design requires.

The page images (`bg1.png` … `bg5a.png`) are **decorative only**: the university crest and plain ruled table backgrounds. No mathematical content exists in them.

B.4 Figures kept as printed, with a recomputation in a note

#	Module page	As printed	Recomputed
1	14	Relative efficiency defined as $\mathrm{Var}_{srs}/\mathrm{Var}_{cl}$ and also called "the design effect"; intra-class form given as $1/[1+(M-1)\rho]$	both forms given and the two definitions distinguished
2	18	Guava example: intermediates printed as detached symbols (4.119, 70.1, 117.6, 1686.6, 297.48, 227.9)	from the sixty printed yields: $\bar{y}=20.00$, $s_b^2=100.28$, $v(\bar{y})=5.71$, s.e. $=2.39$, $E=0.61$ — conclusion unchanged
3	18–19	Sheep assessment answer printed as symbols only	from the table: division means 227.50, 262.375, 409.875, 215.00; $\bar{Y}=222,950$; s.e. $=35,168$
4	21	Lesson divider prints "L WO MULTISTAGE CLUSTER / E SAMPLING. / S / S / O / N / T" (LESSON TWO set vertically)	set as a heading
5	26–28	Wheat example: field sums of squares printed as scattered symbols; final comparison 0.948	$v(\bar{y})=0.00846$, s.e. $=0.092$, $E=0.94$ — same conclusion
6	30	Restaurant activity answer printed as "(5.45, 0.281)"	$\bar{y}=5.46$; two-stage s.e. $=0.166$; the printed 0.281 is the single-stage value using the element variance

#	Module page	As printed	Recomputed
7	46–47	Paddy example: intermediate figures detached (“15.6”, “77.5”, “13.7.0”, “136.0”, “178.1”, “194.0”)	recomputed in full from the printed totals: $\bar{y}_{dreg} = 14.70$, s.e. $= 0.268$; $\bar{y}_{dratio} = 14.75$, s.e. $= 0.269$; $v_{srs} = 0.1392$; $E \approx 1.9$ — module’s conclusion “the two designs are equally efficient” reproduced
8	82–83	Milk Example 4.1: means 414.8 and 471.2, “average change $= 56.4$”	from the printed yields: $\bar{y}_1 = 412.7$, $\bar{y}_{2m} = 407.57$, $\bar{y}_{2u} = 525.0$, change $= 30.1$; the module’s variance components and its “289% more efficient” conclusion are reproduced
9	67, 71	Lesson 4 opens “4.1Introduction”; the variance at $t = 2$ is given without stating its two cases	numbered 4.1; cases (a) and (b) separated
10	78, 81	“4.3 Assessment” is followed by “4.3References” and then by “Example 4.1” out of order	the assessment and references are kept in order, with the worked Example 4.1 placed after them
11	83, 85	“1.1.1 Types and Sources of errors” in Lesson 5; imputation lists numbered iv) and v) under (b)	renumbered 5.1.1, and (i)/(ii)
12	2, 4	Flow chart and course-requirements run-on text (“WEEK 1 & 2: … CAT1 CAT2”)	rebuilt as a week-by-week table; stray “A” dropped
13	9–12, 22–23, 68–70	Notation and definition lists printed without their symbols (e.g. “Mean per unit of the *i*th cluster.” alone on a line)	each definition restored with its symbol

Three further entries in the earlier, complete-record edition (printed pages 50–51, 53 and 59) concerned the foreign SST 404 section. That section is no longer carried here, so those entries are recorded in the SST 404 edition instead.

B.5 What was not altered

The prose is the module’s own wording, including its usage and spacing quirks (“The means of cluster means”, “1.2.1-E-Tivity”, “Regresion” in the Lesson 3 title), except where a typographical slip made a sentence unreadable. Each lesson’s reference list is printed once where the module prints it.

APPENDIX C — SOURCE PAGE MAP

See the table in Appendix A, which maps all 90 file pages to the sections of this edition.